**Local Bank**

Secure • Simple • Together

# Customer 360 & Operations Platform

## User Manual — how to use the application

For branch and head-office employees: tellers, officers, checkers, managers, accountants, auditors and administrators.

Version 1.0 · 09 October 2026

Internal use only. Screens shown use sample data, not real customers.

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# 1. About this application

The **Local Bank Customer 360 & Operations Platform** is the employee system for running branch operations. It brings everything about a customer onto one screen — accounts, loans, fixed deposits, transactions, KYC documents, requests and complaints — and lets staff carry out counter work safely.

It is an **employee-only** application. Customers never sign in to it; every action is carried out by a member of staff on the customer's behalf, and every action is recorded in the audit log.

## What you can do with it

| Area | What it covers |
|---|---|
| Dashboard | Branch figures at a glance: deposits, loans, approvals, KYC, alerts and trends. |
| Customer 360 | Find any customer and see their full relationship; start counter tasks from one place. |
| Accounts & transactions | Open, freeze and close accounts; post cash deposits and withdrawals. |
| Loans | Personal, home, vehicle, business, education and gold loans: application, approval, disbursal, EMI collection. |
| Deposits & FD | Book fixed deposits, early closure, automatic maturity payout. |
| KYC & documents | Collect and verify identity documents; track expiry. |
| Approvals | Maker-checker: every sensitive change is reviewed and approved by different people. |
| Tickets | Customer complaints and service requests, tracked to resolution. |
| Alerts | Fraud and risk alerts, overdue loans, expiring KYC, maturing FDs. |
| Reports | 14 standard reports, on screen or exported to Excel / PDF. |
| Accounting | Double-entry ledger, trial balance, reconciliation, savings interest posting. |
| Administration | Employees, roles and permissions, branches, settings and the audit log. |

## Key ideas to understand first

- **Roles decide what you see.** The menu only shows the areas your role can use. If a button is missing, your role does not have that right — the server also refuses the action, so it cannot be worked around.
- **Maker-checker.** Opening an account, sanctioning a loan, booking an FD, high-value transactions and other sensitive changes are *requests*. A different employee reviews them and a manager approves them before they take effect.
- **Branch scope.** Most employees see only their own branch. Head-office roles see all branches.
- **Everything is logged.** Searches, views of sensitive data, changes and approvals are written to a tamper-evident audit log.
- **Money is exact.** Amounts are kept to the paisa and rounded half-up; balances can never go negative.

# 2. Getting started

## Opening the application

Your administrator will give you the address of the application. On the branch computer that runs it, it is usually **http://127.0.0.1:5000**. Use a current version of Microsoft Edge, Google Chrome or Firefox.

> **Tip:** For administrators — starting the application on the server computer: open a terminal in the application folder and run `docker compose up -d` (starts the database), then `python run.py`. The first start brings the database up to date automatically. See chapter 16 for details.

## Your first sign-in (invite link)

New employees do not receive a password. Your administrator creates your account and gives you a **one-time invite link**.

1. Open the invite link in your browser. It works **once** and expires after **48 hours**.
2. Choose your password and type it twice. It must be at least **12 characters**, mix letters, numbers and symbols, and must not contain your name or common words.
3. Select **Set password**. You are told "Your password is set". Now sign in normally.

> **Important:** If the link has expired or was already used, ask your administrator for a new one. Never forward an invite link to anyone else.

## Signing in

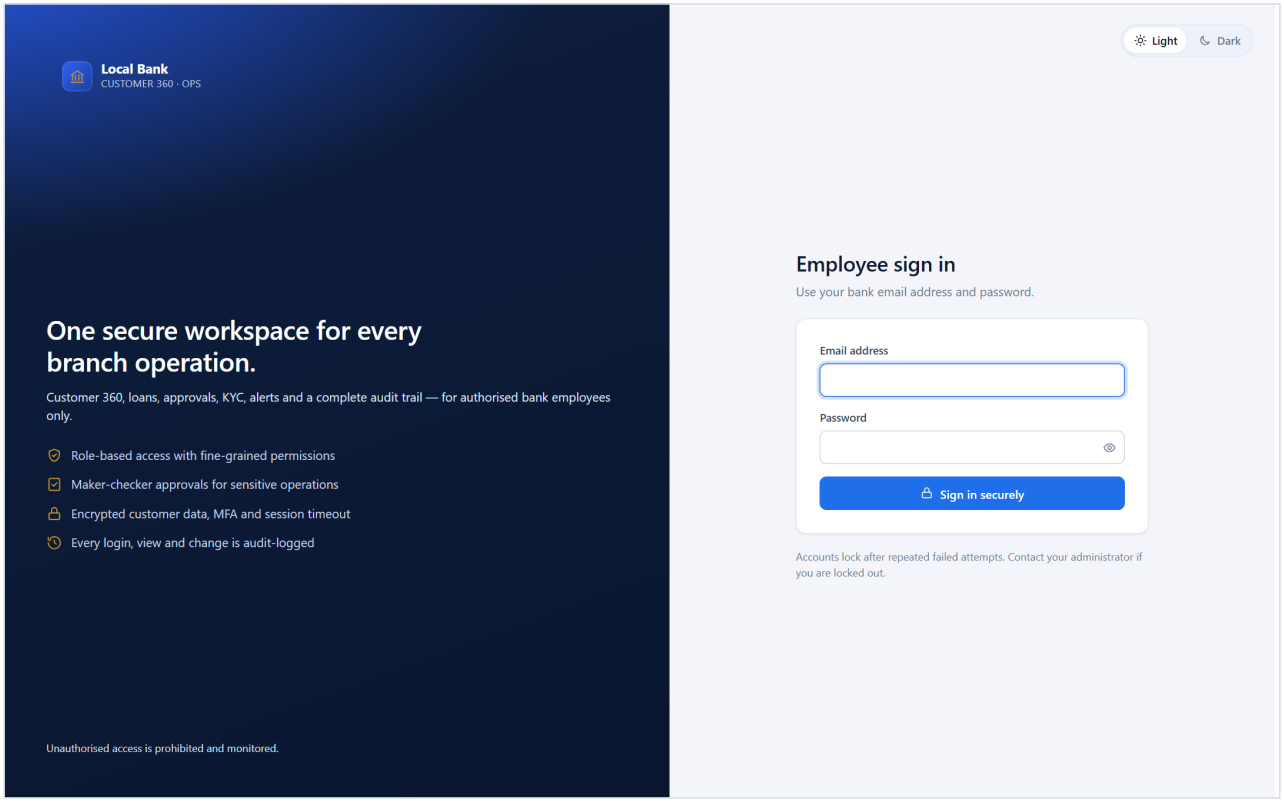



*Figure 2.1 — The sign-in screen.*

1. Enter your work email address and password, then select **Sign in**.

2. If two-factor authentication is on for your account, enter the 6-digit code from your authenticator app.
3. You arrive on the Dashboard.

For your protection:

- After **5 wrong passwords** your account is locked for **15 minutes**. Too many attempts from one computer are also slowed down.
- You are signed out after **15 minutes without activity**, and after **12 hours** in any case.
- Passwords expire every **90 days**; the application asks you to choose a new one.
- The error message never says whether the email or the password was wrong — this is deliberate.

## Two-factor authentication (recommended)

1. Open the menu under your name (top right) and choose **Two-factor authentication**.
2. Scan the QR code with an authenticator app (Microsoft Authenticator, Google Authenticator or similar).
3. Type the 6-digit code shown in the app and select **Enable MFA**.
4. From now on, sign-in asks for a fresh code each time. Each code can be used only once.

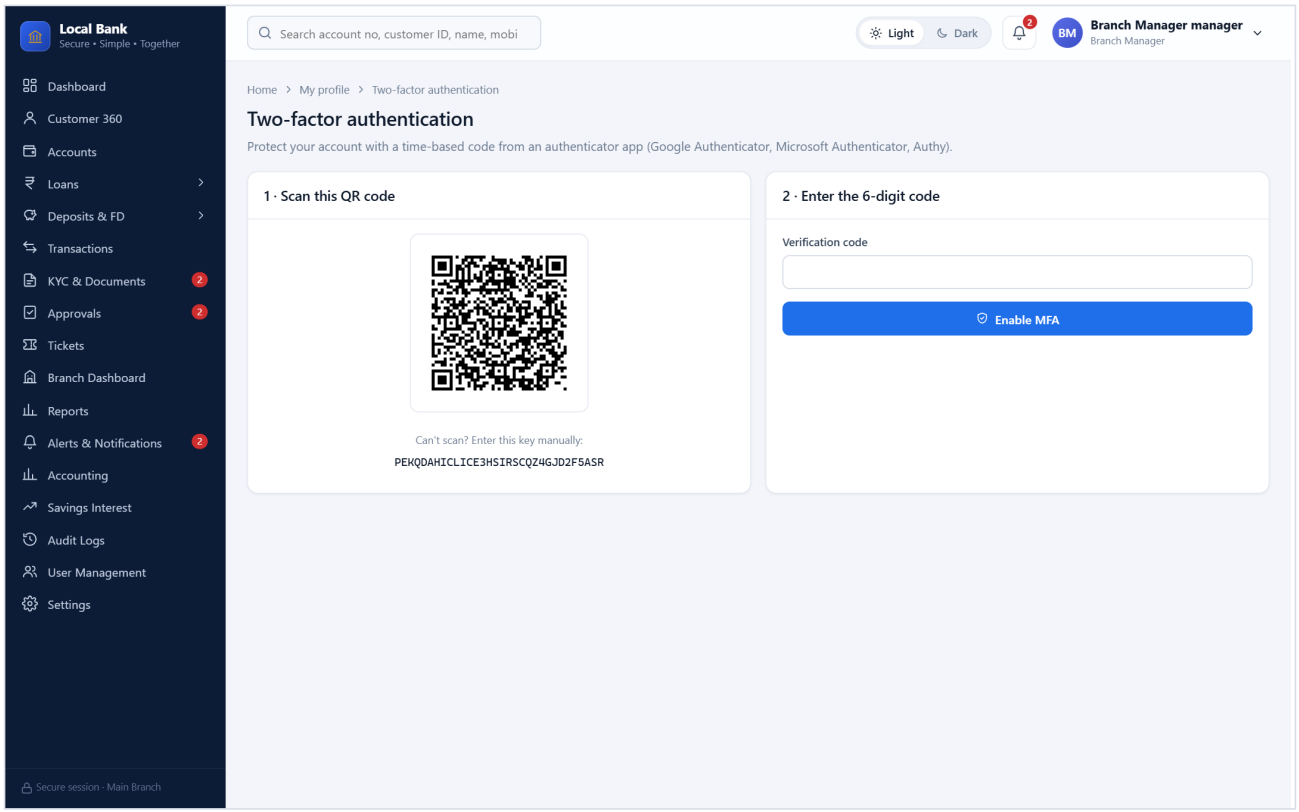



*Figure 2.2 — Setting up two-factor authentication.*

## Changing your password or signing out

Use the menu under your name: **My profile**, **Change password**, **Two-factor authentication** and **Sign out**. Always sign out when you leave the counter.

## Forgot your password?

Ask your administrator to send you a **password reset link**. It works like an invite link. Your old password stops working immediately and any open sessions are signed out.

# 3. Finding your way around

## The screen layout

- **Left menu** — the areas your role can use. A red number next to **KYC & Documents** or **Approvals** is the number of items **waiting for you**. The number next to **Alerts** is your unread alerts.
- **Search bar** (top) — find any customer, account or loan from any screen.
- **Light / Dark** — switch the colour theme. Your choice is remembered on this computer.
- **Bell** — opens your alerts.
- **Your name** — profile, password, two-factor authentication and sign out.

## Searching

Type into the search bar and press **Enter**. You can search by:

| Search by | How it matches | Example |
|---|---|---|
| Account number | Exact | 504953498837 |
| Customer ID | Exact | CIF392382 |
| Loan number | Exact — opens the loan directly | LNPE51428916 |
| Mobile number | Exact (stored encrypted) | 9840011122 |
| PAN | Exact (stored encrypted) | PRIYA1234K |
| Name | Partial, at least 3 letters | Priya |

If exactly one customer matches, the application opens their **Customer 360** straight away. If several match, you get a list — select the name to open it.

# Light and dark theme

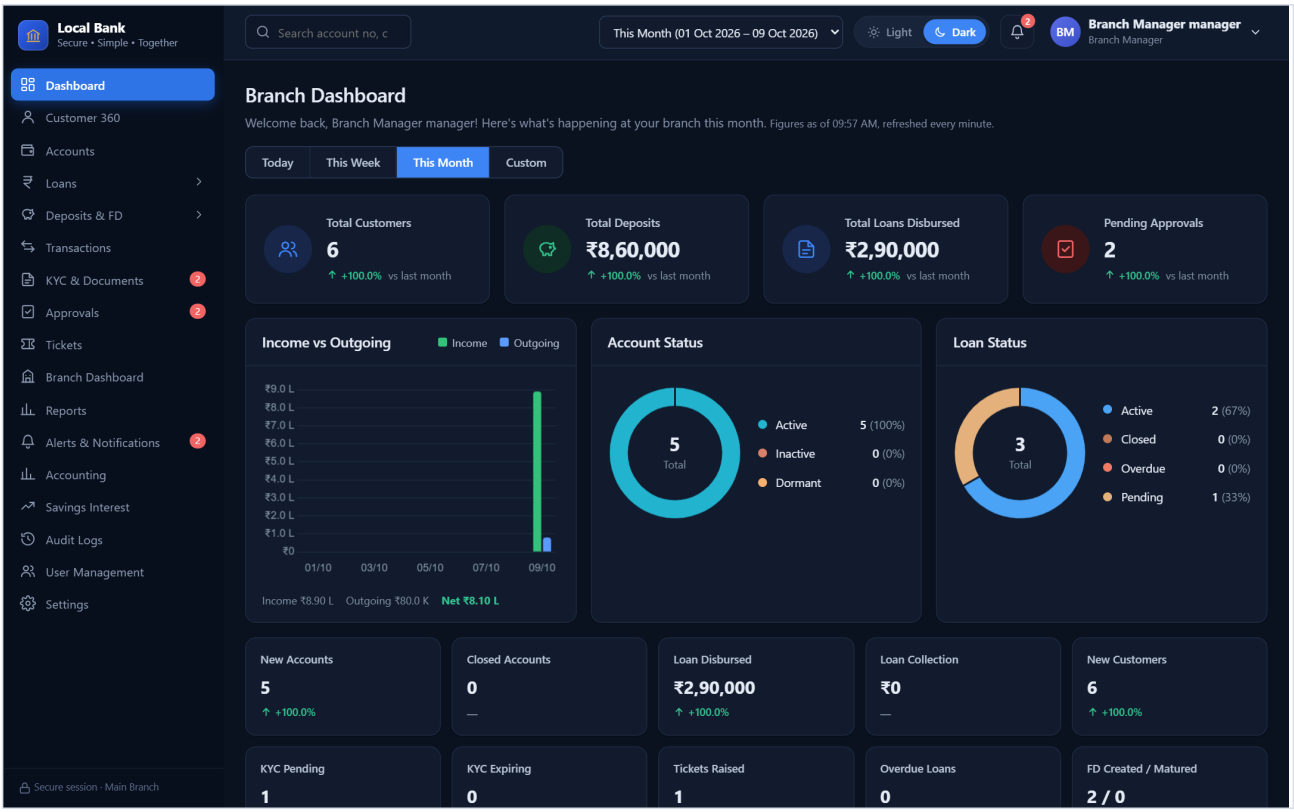



*Figure 3.1 — The same dashboard in the dark theme.*

# Confirmation dialogs

Before anything that moves money or changes a record, the application shows a confirmation that states exactly what will happen — for example the amount, the account, the customer and the current balance. **Read it before you confirm.** If anything is wrong, select **Cancel** and correct the form.

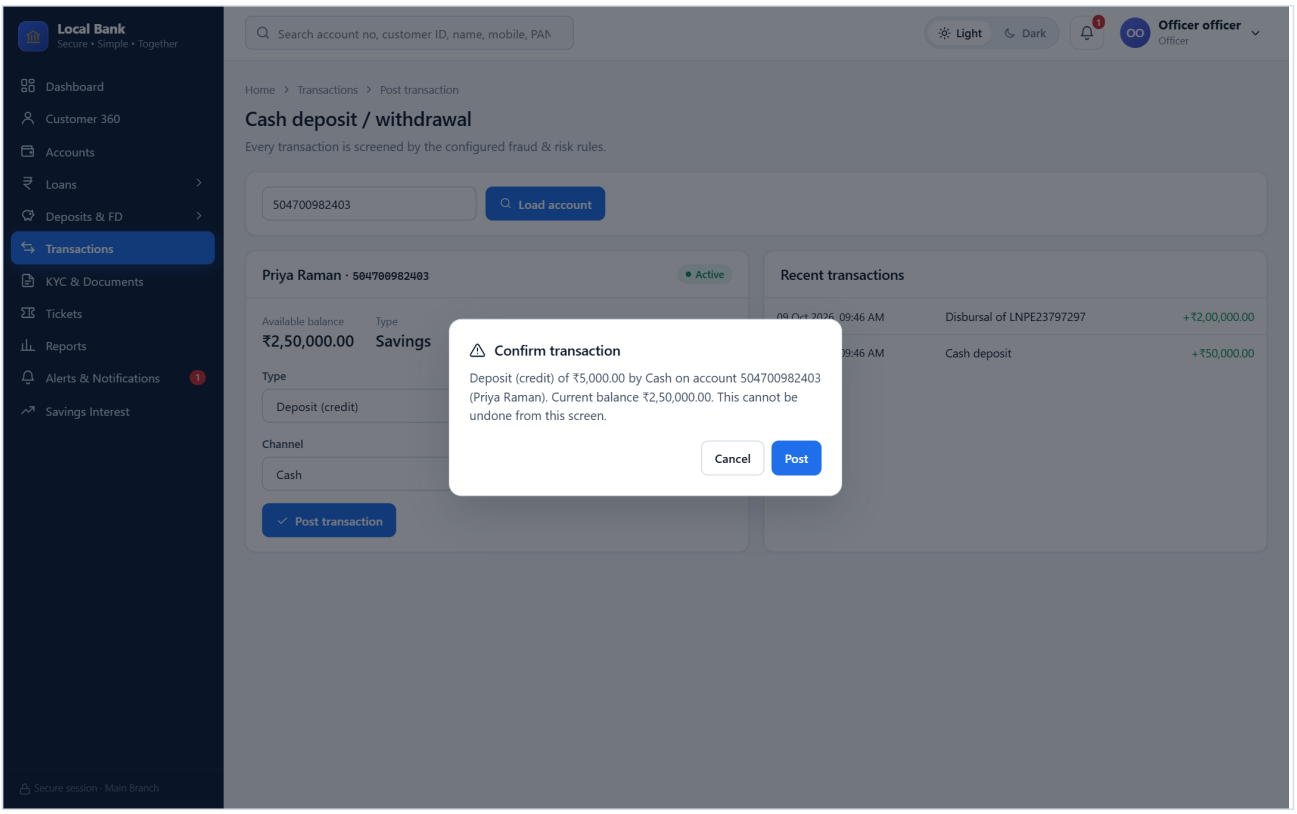



*Figure 3.2 — A confirmation always repeats the amount, type, account and customer.*

# 4. Dashboard

The dashboard is the first screen after sign-in. It summarises your branch (or all branches, for head-office roles).

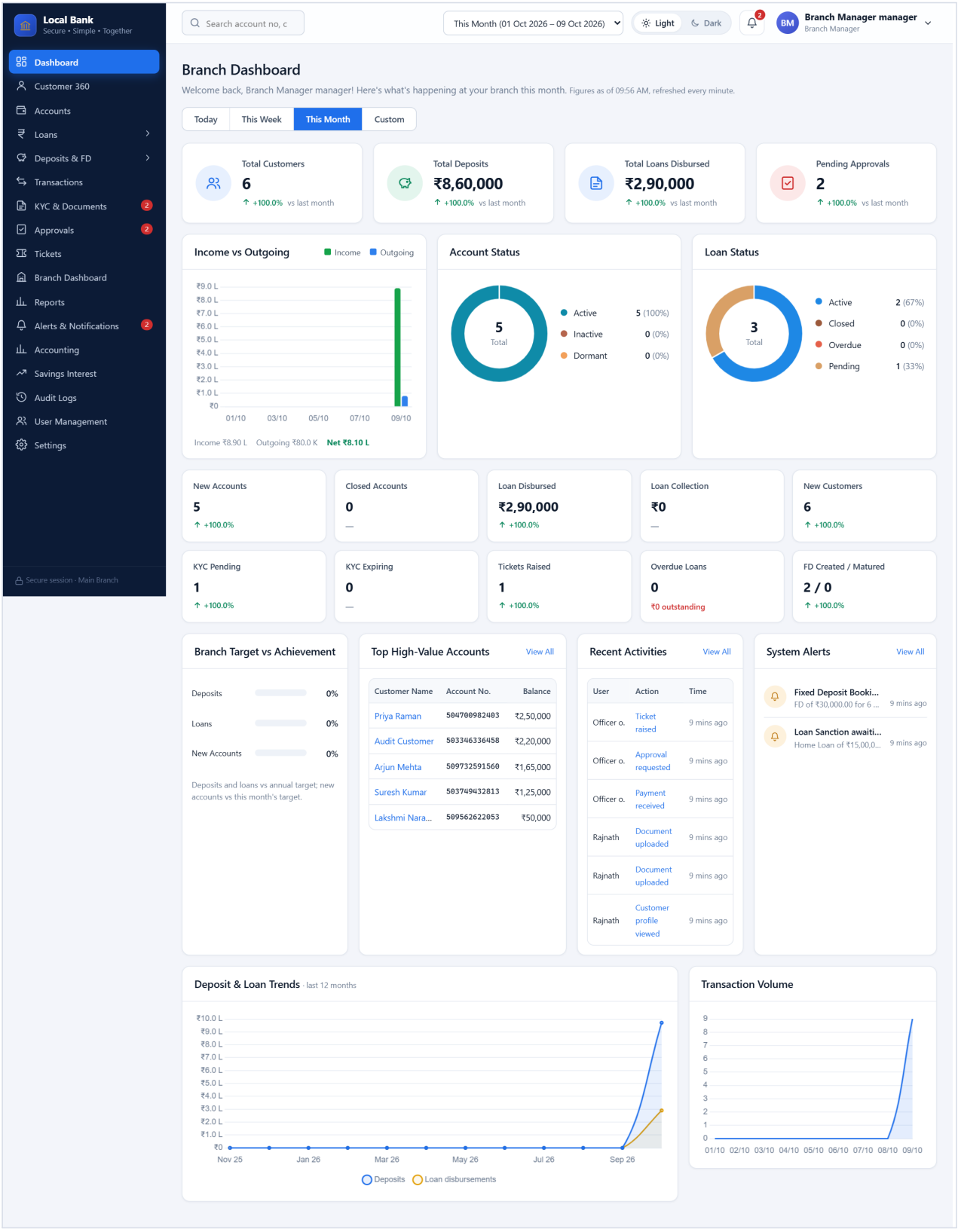



*Figure 4.1 — Branch dashboard.*

- **Period buttons** — Today, This Week, This Month or a Custom range. The arrows show change against the previous period.

- **Top cards** — total customers, total deposits (savings/current balances plus active FDs), loans disbursed and pending approvals.
- **Charts** — income vs outgoing, account and loan status, 12-month trends and transaction volume.
- **Lower cards** — new and closed accounts, collections, KYC pending and expiring, tickets, overdue loans and FDs.
- **Tables** — branch targets, top accounts, recent activity and system alerts. Select a name to open the customer.

Figures are refreshed every minute; the time they were calculated is shown under the title. Managers also have a **Branch Dashboard** in the menu with targets and branch performance.

# 5. Customer 360

Customer 360 is where most counter work starts. It shows everything the bank holds for one customer.

## Finding a customer

1. Select **Customer 360** in the menu, or use the search bar at the top.
2. Type an account number, customer ID, name, mobile or PAN and press Enter.
3. Select the customer's name (or **Open 360**).

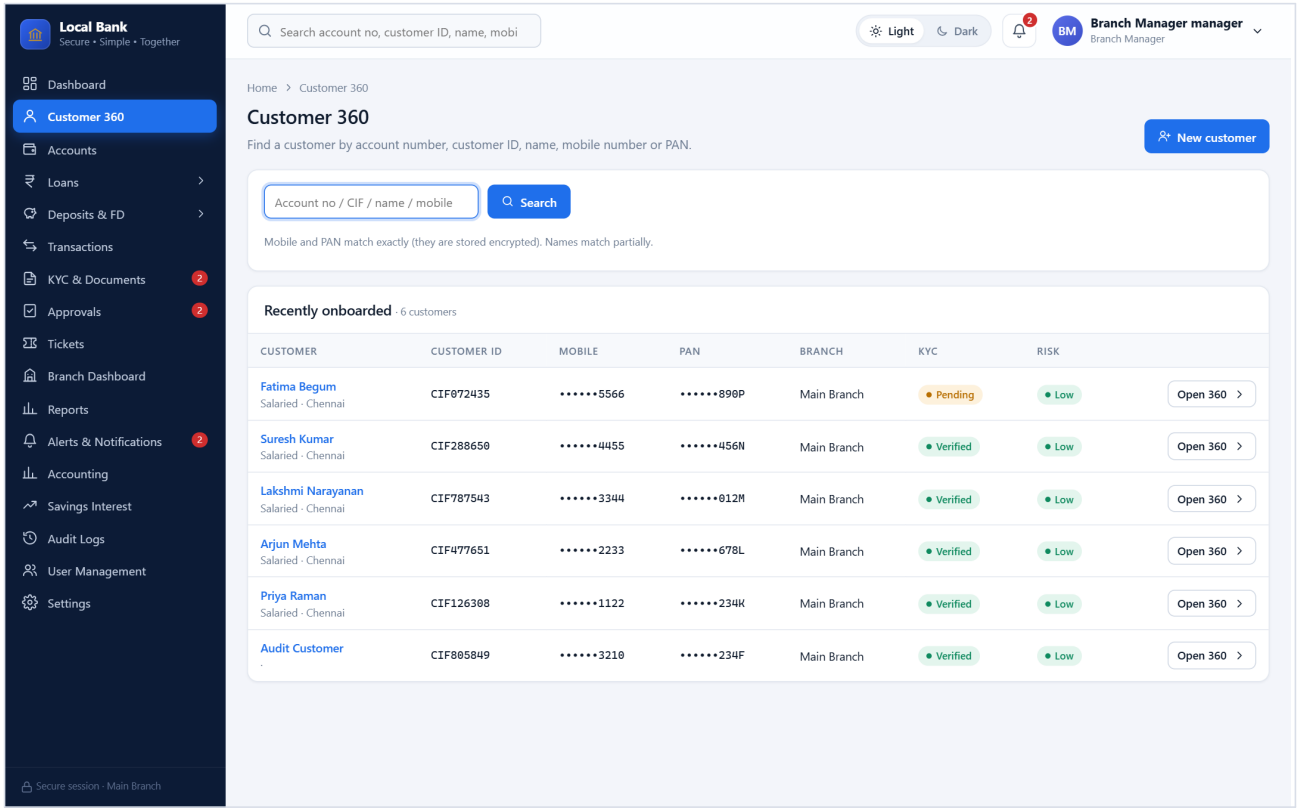



*Figure 5.1 — Customer search; recently onboarded customers are listed when nothing is typed.*

# Reading the Customer 360 screen

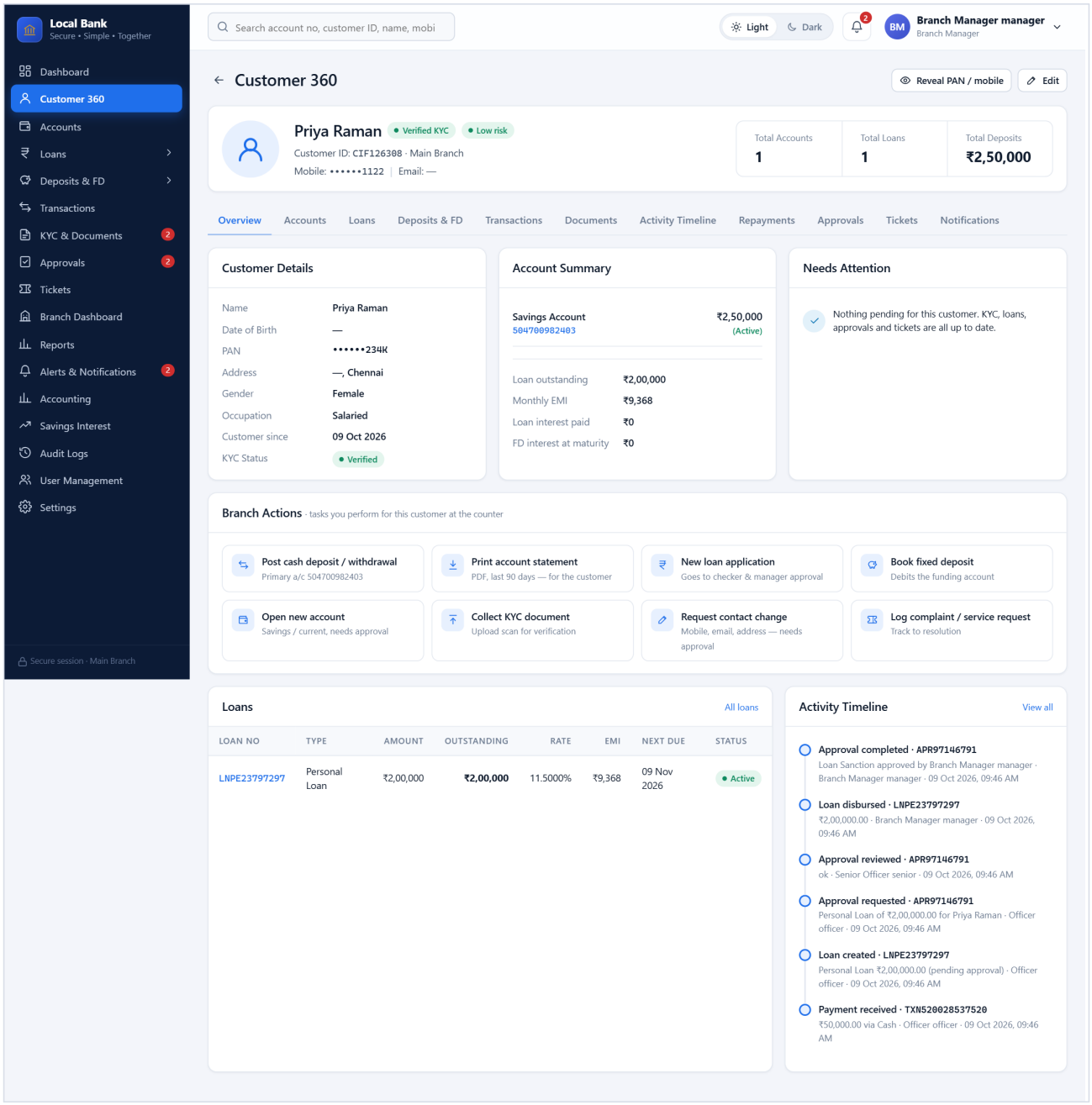



*Figure 5.2 — Customer 360 overview.*

- **Header** — name, KYC status, risk category, customer ID, home branch and masked mobile number, with totals.
- **Tabs** — Overview, Accounts, Loans, Deposits & FD, Transactions, Documents, Activity Timeline, Repayments, Approvals, Tickets and Notifications. You only see tabs your role allows.
- **Needs Attention** — anything pending for this customer (KYC, overdue EMI, approvals, open tickets) with a button to act.
- **Branch Actions** — the counter tasks: post a deposit/withdrawal, print a statement, new loan, book FD, open account, collect KYC, request a contact change and log a complaint.
- **Activity Timeline** — everything that happened to this customer, newest first.

## Seeing the full PAN or mobile number

PAN and mobile are masked (••••••234K). If your role allows, select **Reveal PAN / mobile** and confirm. Every reveal is recorded in the audit log with your name — only reveal when you need to.

## Adding a new customer

1. Go to **Customer 360** and select **New customer**.
2. Enter name, PAN, mobile, branch and the other details. A PAN can belong to only one customer.
3. Select **Create customer**. The customer is created with KYC status **Pending**.
4. Continue with KYC (chapter 9): upload PAN and Aadhaar so a checker can verify them.

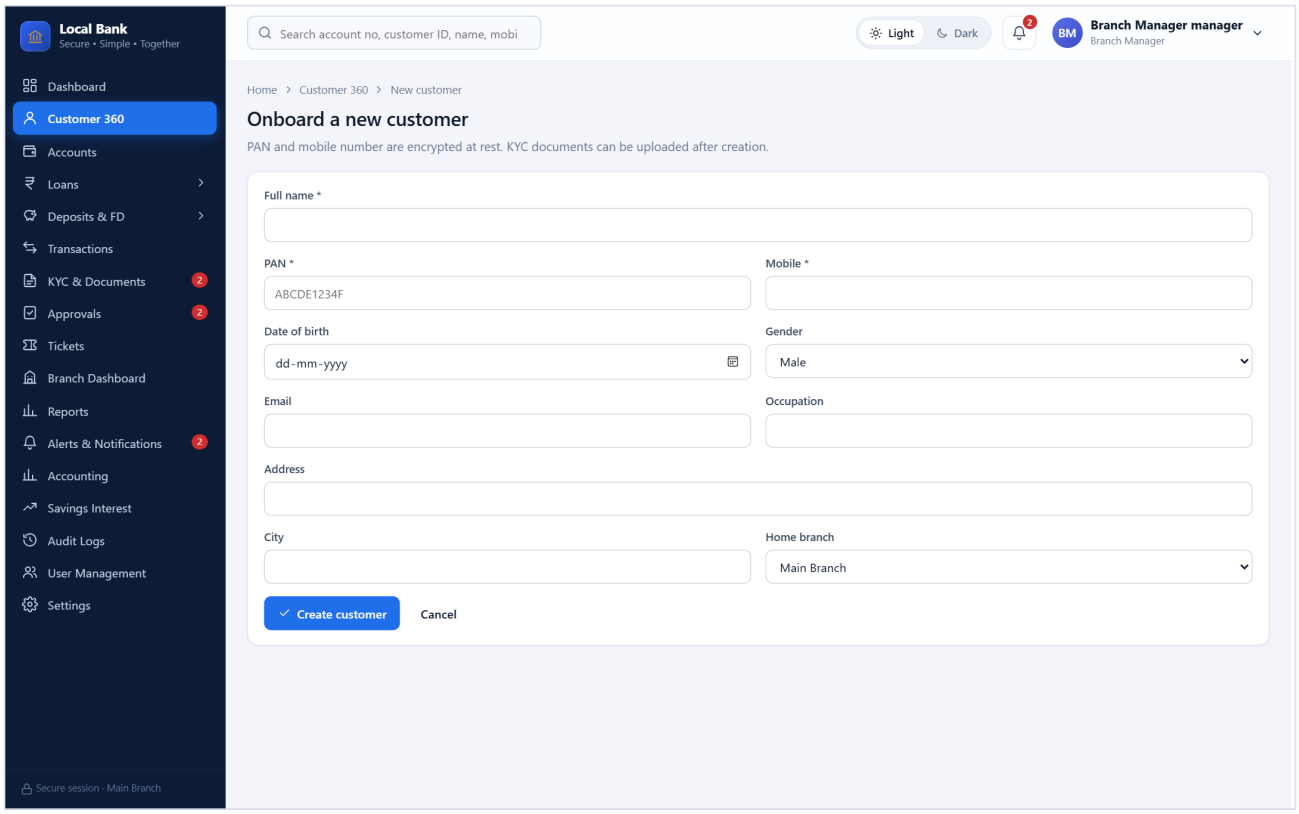



*Figure 5.3 — New customer form.*

## Changing a customer's contact details

Select **Edit** (or **Request contact change**). Changes to mobile, email, address, city, occupation or risk category are sent for **approval**; the old details stay in place until a checker and a manager approve. The request is shown on the customer's Approvals tab.

## Printing an account statement

Select **Print account statement** in Branch Actions. A PDF of the last 90 days opens for the customer. This needs the *print* right on transactions.

# 6. Accounts

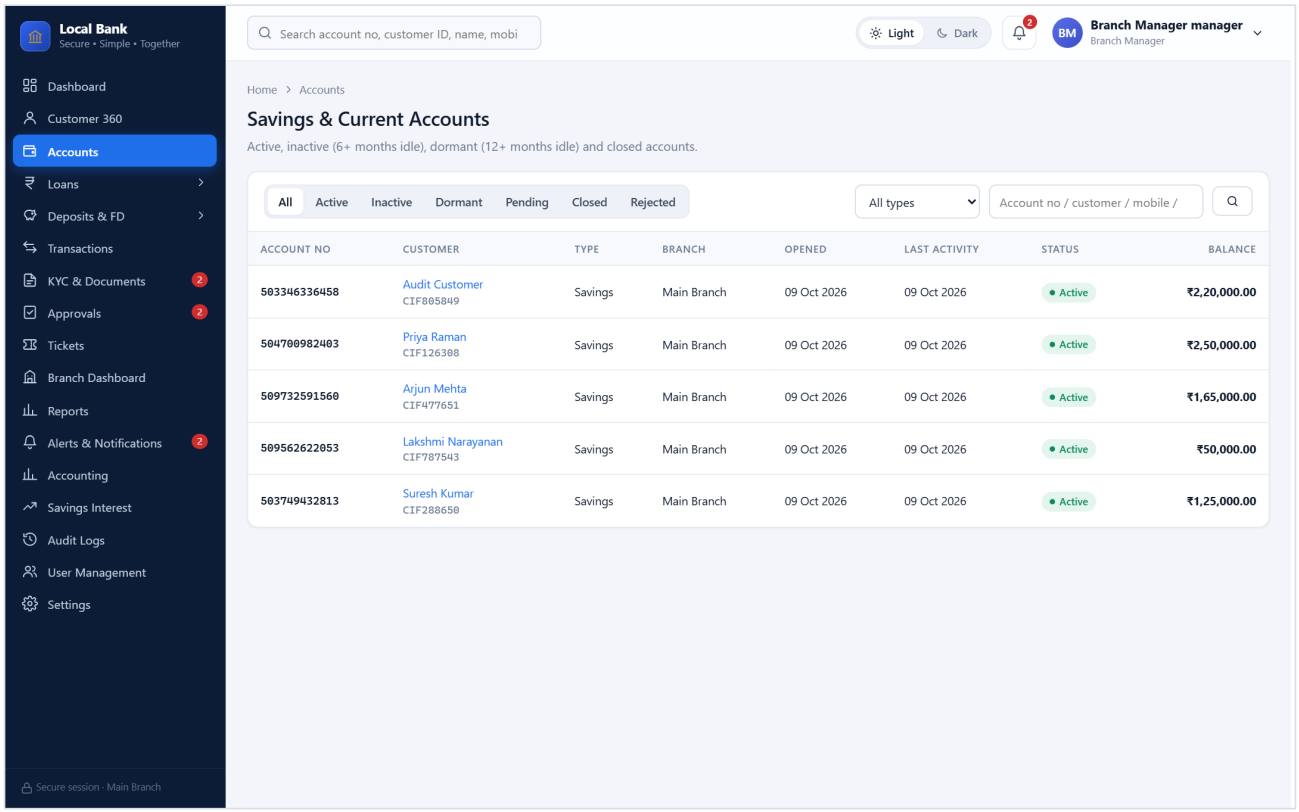



*Figure 6.1 — Account list with filters.*

## Opening a new account

> **Important:** An account can be opened only for a customer whose KYC is **Verified** (PAN and Aadhaar both verified). If KYC is not complete, the form explains what is missing and links to the documents.

Start from either place:

- **Accounts → Open Account**, then search for the customer by name (3 letters or more), mobile, PAN, customer ID or account number and select them. The list shows each customer's KYC status.
- **Customer 360 → Open new account**, when the customer is already open.

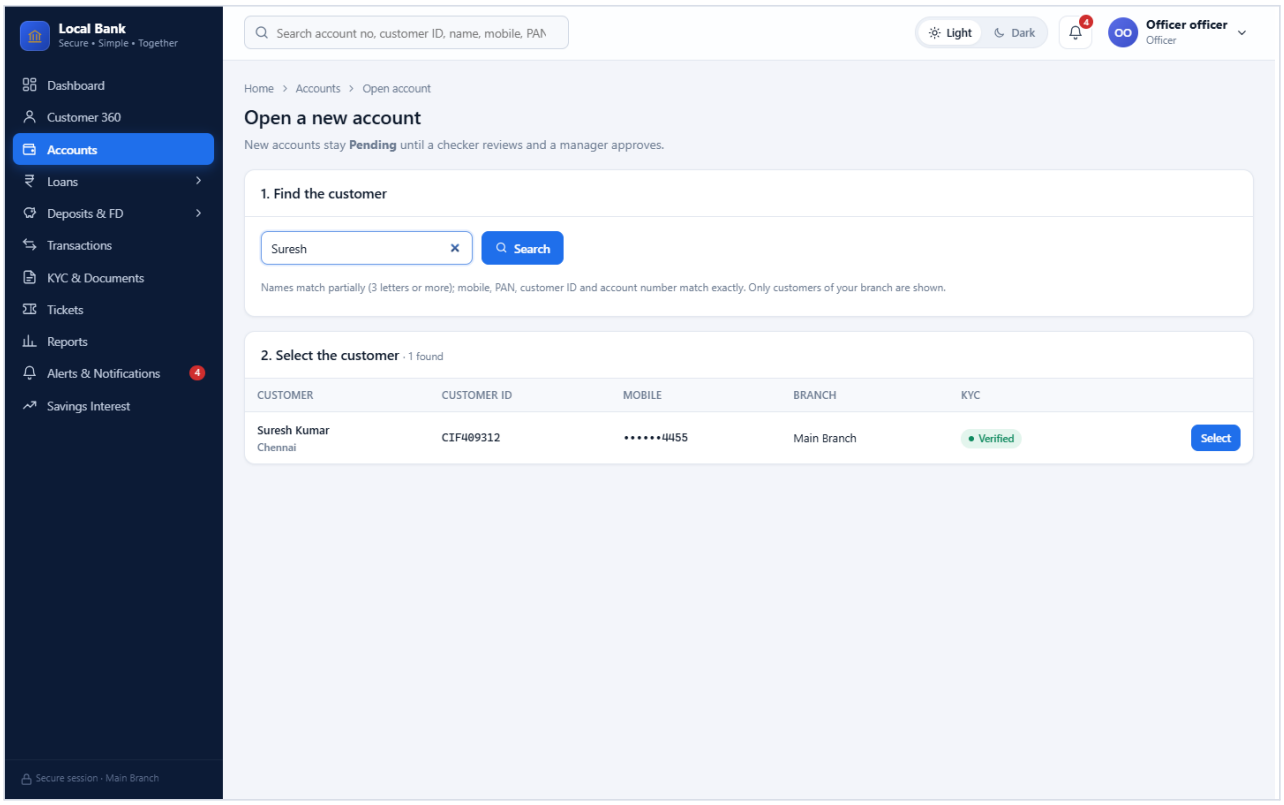



*Figure 6.2 — Open Account: find and select the customer.*

1. Check the customer and their **existing accounts** shown at the top, so you do not open a duplicate. Use **Change customer** if you picked the wrong person.
2. Choose **Savings** or **Current** and select **Submit for approval**, then confirm.
3. The account number is created straight away with status **Pending**. No money can be posted yet.
4. A checker reviews and a manager approves (chapter 10). The account then becomes **Active**.

## Account statuses

| Status | Meaning |
|---|---|
| Pending | Waiting for approval. No transactions allowed. |
| Active | Normal operation. |
| Inactive | No customer transaction for 180 days. Any transaction makes it Active again. |
| Dormant | No customer transaction for 365 days. Any transaction reactivates it and raises a risk alert. |
| Frozen | Deposits are accepted, but withdrawals and debits are blocked (e.g. court order). |
| Closed | Closed after approval. Balance was zero. |
| Rejected | The opening request was rejected. |

## Freezing or unfreezing an account

1. Customer 360 → **Accounts** tab → **Freeze** (or **Unfreeze**) on the account.
2. Enter the reason (required) and select **Send request**.
3. The freeze takes effect when the request is approved.

## Closing an account

The balance must be **₹0** and there must be no active loan or FD linked to it. Select **Close** on the Accounts tab; the closure goes for approval. If the balance is not zero, the application tells you how much is still held.

# 7. Cash deposits and withdrawals

1. Open the customer's Customer 360 and select **Post cash deposit / withdrawal** (or go to **Transactions** → **Post transaction** and load the account number).
2. Check the customer name, account number and available balance shown on the left.
3. Choose **Deposit (credit)** or **Withdrawal (debit)**, enter the amount (rupees and paise), the channel (Cash, Cheque, NEFT, IMPS, UPI, Transfer) and an optional narration.
4. Select **Post transaction**. Read the confirmation — it repeats the type, amount, channel, account, customer and current balance.
5. Select **Post**. The new balance and the transaction reference appear at the top of the screen.

The screen stays on the same account, so you can post the next transaction for the same customer straight away. The last 8 transactions are listed on the right.

## Rules applied to every transaction

| Rule | What happens |
|---|---|
| Withdrawal more than the balance | Refused — the available balance is shown. |
| Account Pending, Closed or Rejected | Refused. |
| Account Frozen | Deposits allowed; withdrawals refused. |
| Amount ₹10,00,000 or more (default) | Not posted yet. Sent for maker-checker approval as a **High-value Transaction**; posted automatically once approved. |
| Amount ₹5,00,000 or more (default) | Posted, and a fraud & risk alert is raised for review. |
| 5 or more transactions in 30 minutes on one account | Posted, flagged and alerted (velocity rule). |
| Transaction on a Dormant account | Posted, account reactivated, alert raised. |

Limits shown are the defaults; administrators can change them in Settings and Alert rules.

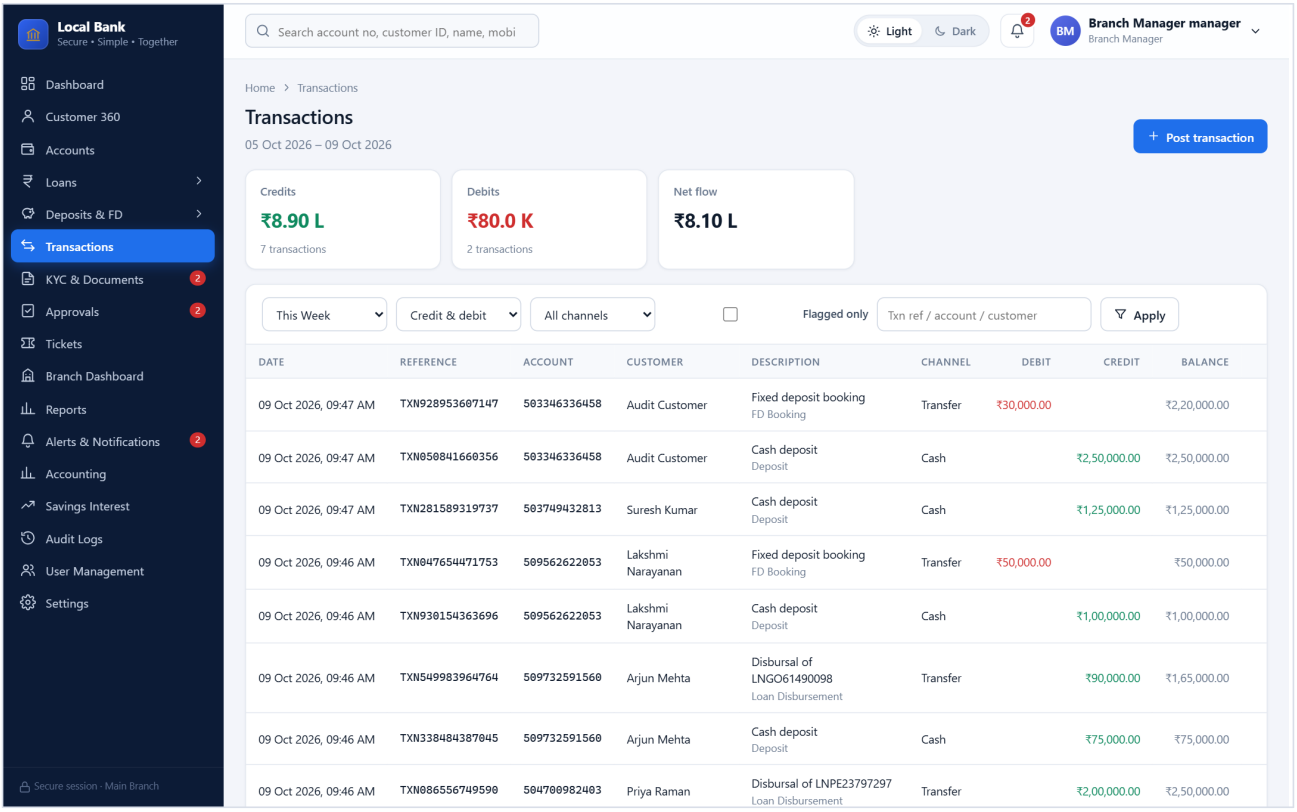



*Figure 7.1 — Transaction list. Flagged transactions are marked.*

> **Important:** A posted transaction can never be edited or deleted — the ledger is append-only. A wrong deposit or withdrawal is corrected with an authorised **reversal** (below).

## Transfers between accounts

Moves money from one account in this bank to another — the customer's own accounts or another customer's — in one step. Both accounts must be in your branch (head-office roles: any branch).

1. **Transactions** → **Transfer**, or Customer 360 → **Transfer between accounts** (the customer's account is filled in).
2. Enter the **from** and **to** account numbers and select **Load accounts**. Check both names and balances.
3. Enter the amount and an optional narration, select **Transfer** and confirm.
4. Both balances update together and two entries are recorded: *Transfer Out* and *Transfer In*.

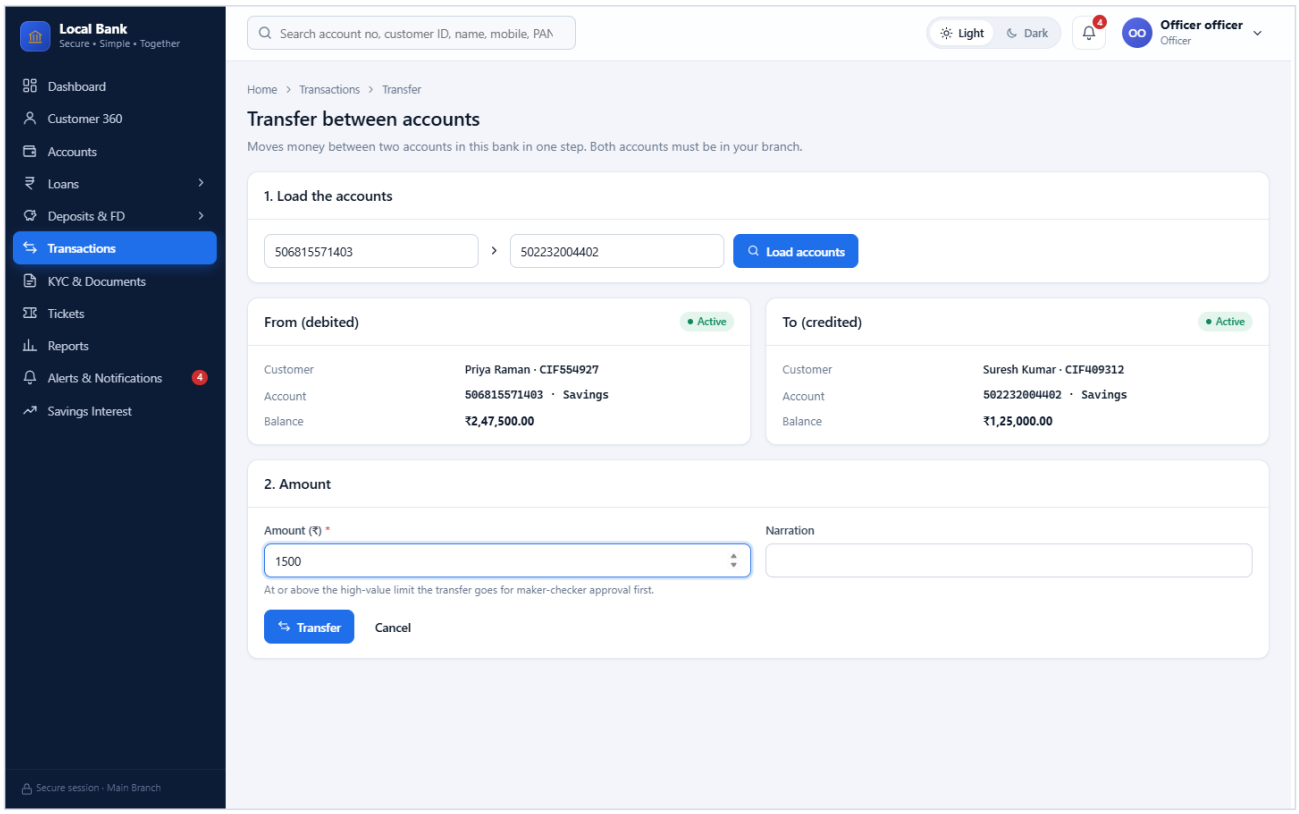



*Figure 7.2 — Transfer between accounts.*

- A frozen account cannot send money; it can receive it.
- At or above the high-value limit (₹10,00,000 by default) the transfer goes for maker-checker approval first.
- If a transfer was wrong, make a transfer back the other way.

## Correcting a mistake: reversal

If a teller deposit or withdrawal was posted wrongly (wrong amount, wrong account), it is reversed through maker-checker:

1. Open the transaction in **Transactions** or Customer 360 → **Transactions** tab and select **Reverse**.
2. Enter the reason (required) and select **Send for approval**. Nothing changes yet.
3. A checker reviews and a manager approves. The opposite entry is posted, the original is marked **Reversed** and the new entry is marked **Reversal**, with the original reference in its description.
4. Then post the correct transaction, if one is needed.

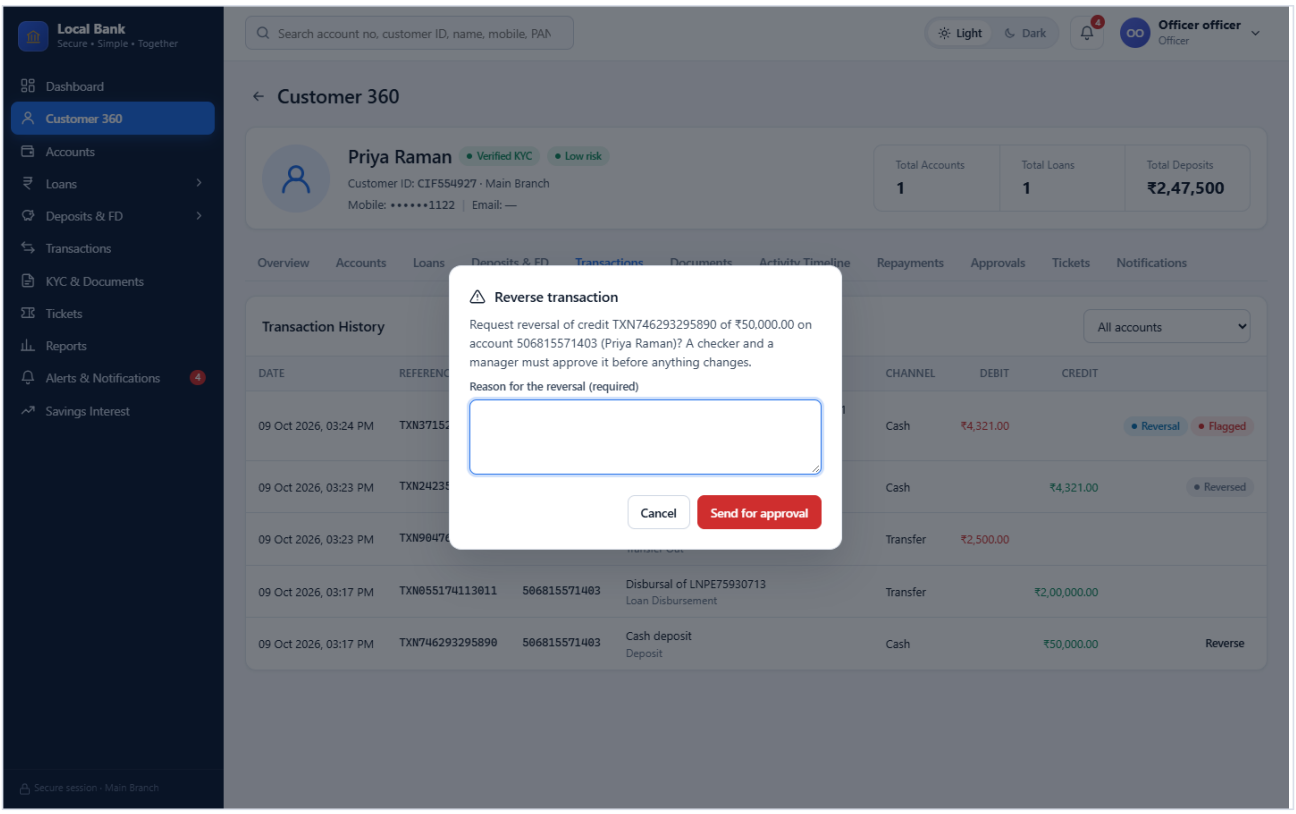



*Figure 7.3 — Requesting a reversal; the reason is required.*

- Only deposits and withdrawals can be reversed here. Loan, FD, interest and transfer postings are corrected through their own process.
- Reversing a deposit is refused if the money has already left the account, or if the account is frozen.
- A transaction can be reversed only once.

# 8. Loans

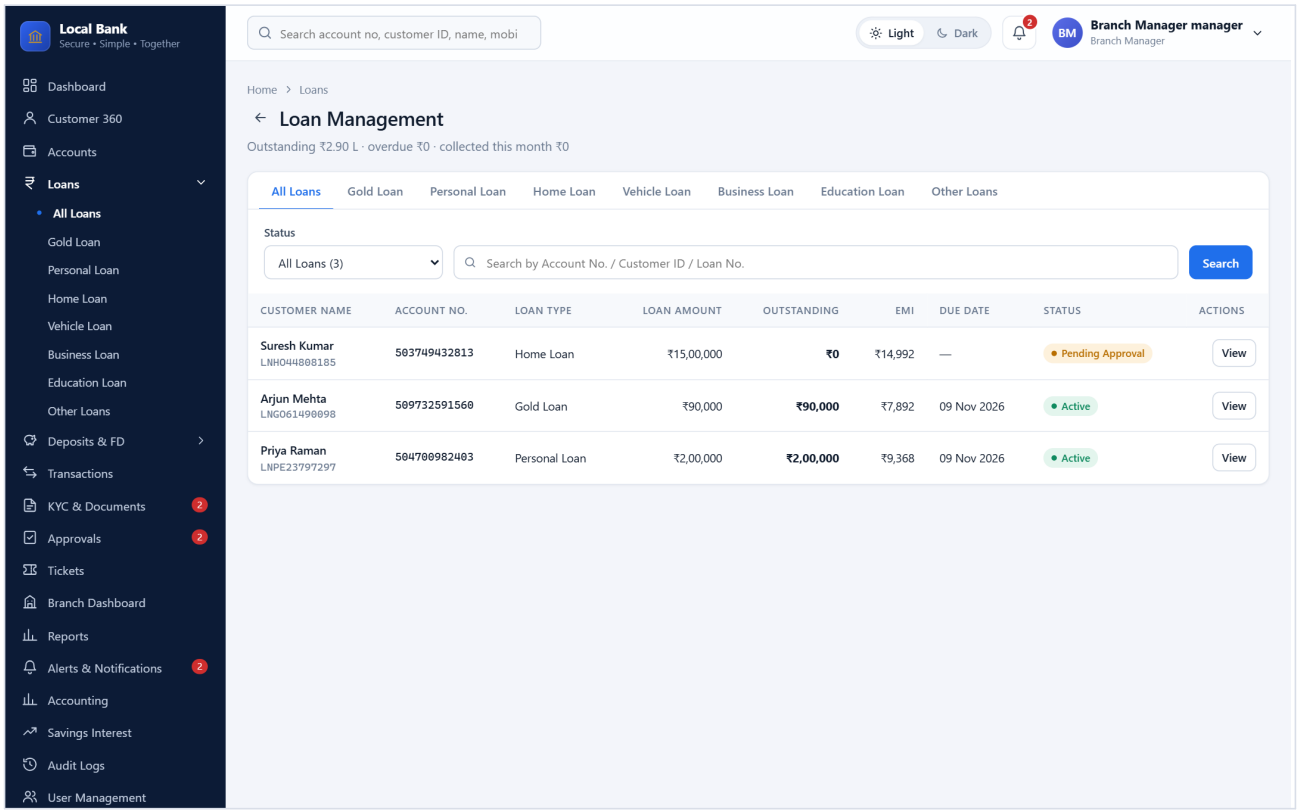



*Figure 8.1 — Loan book. Use the sub-menu for Gold Loan and each loan type.*

## Loan types

Gold, Personal, Home, Vehicle, Business, Education and Other loans. Gold loans have their own screen with weight, purity and valuation.

## Creating a loan application

> **Important:** The customer's KYC must be **Verified**, and they need an active account to receive the money.

1. Open the customer's Customer 360 and select **New loan application**.
2. Choose the loan type and the disbursement account.
3. Enter amount (₹1,000 to ₹100 crore), interest rate (above 0% and up to 36% a year) and tenure (1 to 360 months). The estimated EMI is calculated as you type.
4. For a **gold loan**, also enter gold weight (grams), purity and valuation. The loan cannot exceed **75%** of the valuation.
5. Select **Submit**. The loan is created as **Pending** and a **Loan Sanction** request goes for approval.
6. When approved, the loan amount is credited to the disbursement account and the loan becomes **Active**.

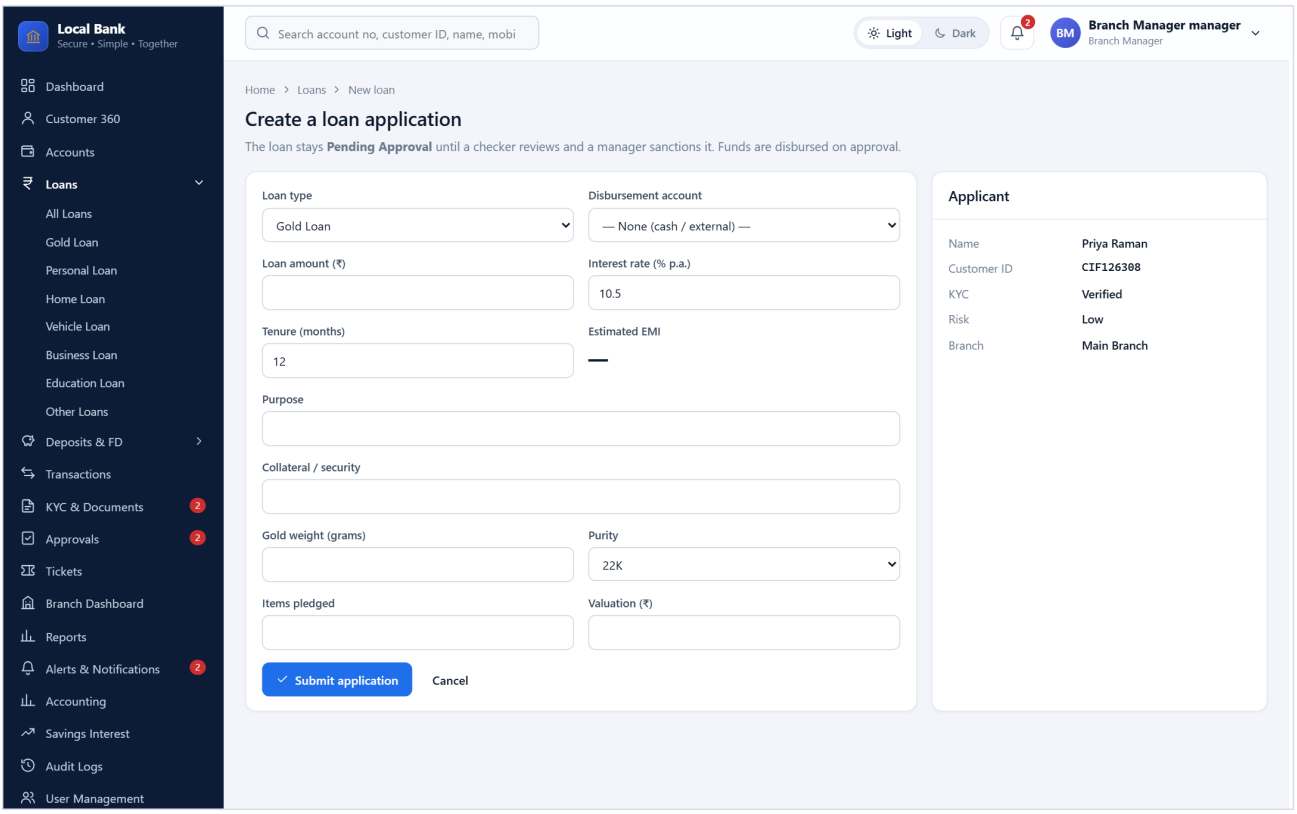



*Figure 8.2 — New loan application with EMI estimate.*

# The loan page

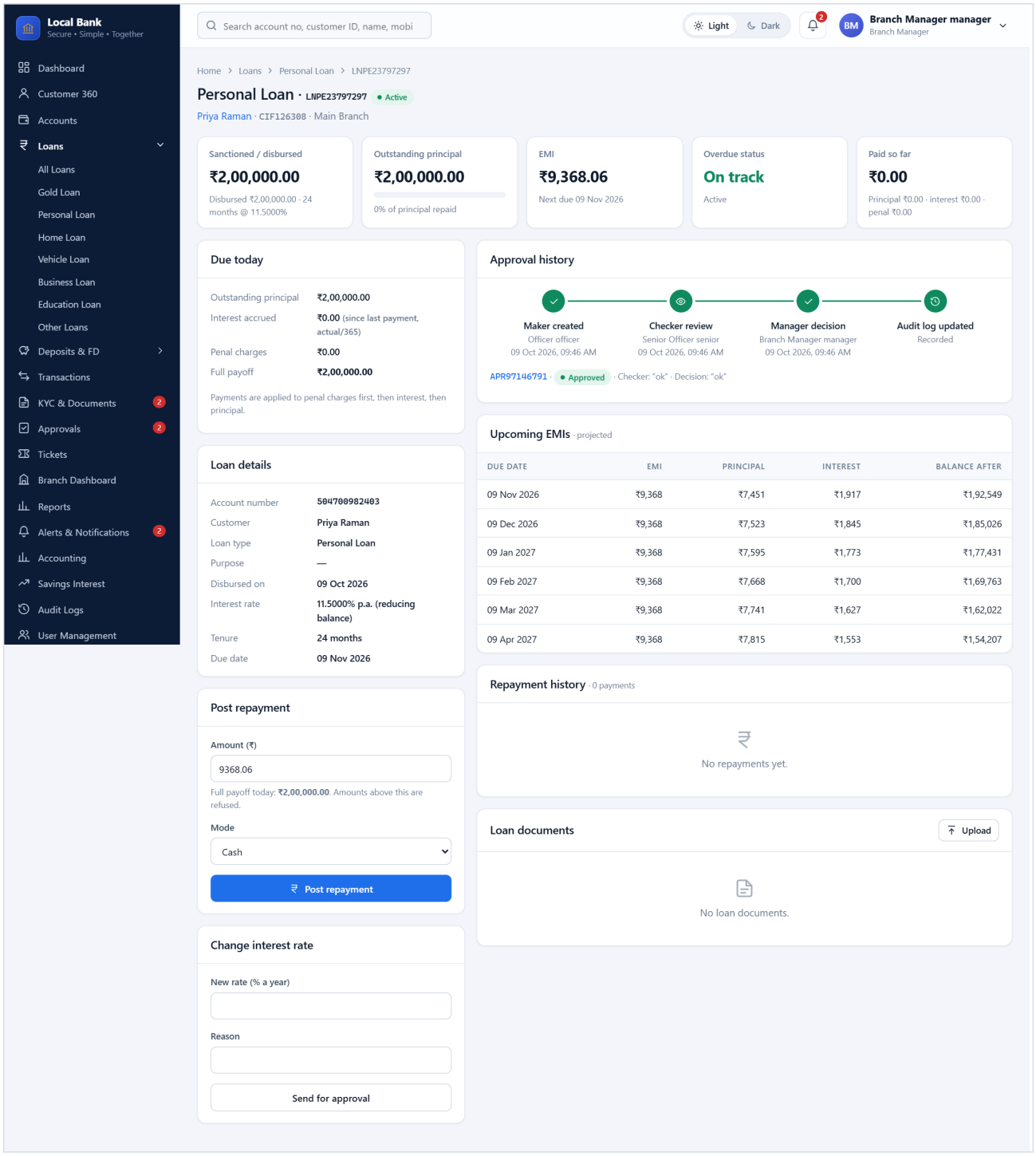



*Figure 8.3 — Loan details: dues, repayment, schedule and documents.*

The loan page shows sanctioned and disbursed amounts, outstanding principal, EMI, next due date, the amount due today (interest and any penal charge), the full payoff amount, the repayment schedule and loan documents.

## Collecting an EMI or repayment

Repayments are posted by roles with the loan *edit* right: Loan Officer, Senior Officer and Branch Manager. Other roles see the amount due and who can collect it.

1. Open the loan (search the loan number, or Customer 360 → Loans).
2. In **Post repayment**, the amount due now is filled in. Change it if the customer pays a different amount (it cannot be more than the payoff amount).
3. Choose the mode: Cash, NEFT, UPI, Cheque or Account debit (from the linked account).
4. Select **Post repayment**, read the confirmation and select **Post repayment** again.

Every payment is applied in this order: **penal charges → interest → principal**. Interest is simple interest on the outstanding principal for the actual days since the last payment (actual/365). Paying the full payoff amount closes the loan.

## Overdue loans and penal charges

- The scheduled scan marks a loan **Overdue** when the due date passes and counts the days overdue.
- A penal charge of **24% a year** (default) is charged on the overdue instalment for the days it is late.
- Alerts are raised for EMIs due soon (5 days, default) and overdue loans; after 60 days overdue the alert is High severity.

## Changing a loan's interest rate

On the loan page, enter the new rate and a reason under **Change interest rate** and select **Send for approval**. When approved, the EMI is recalculated for the remaining tenure.

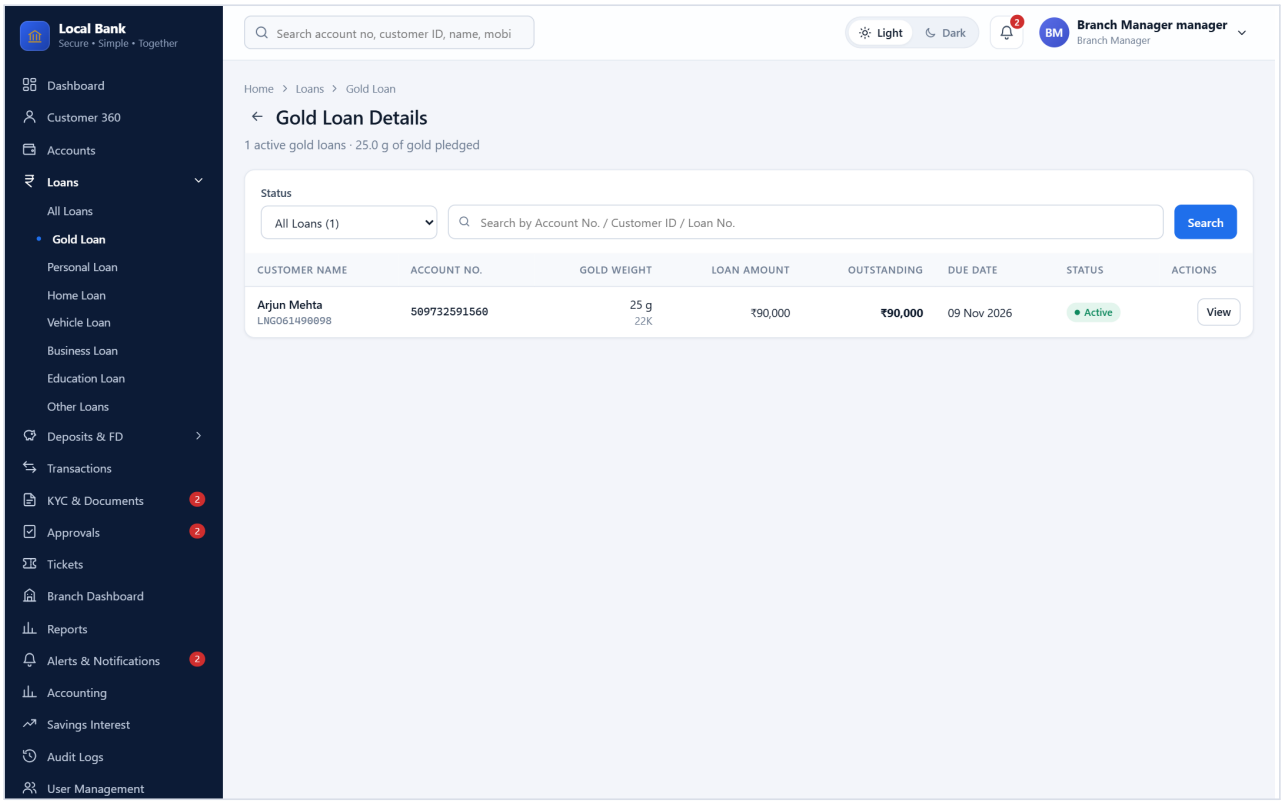



*Figure 8.4 — Gold loan portfolio.*

# 9. Fixed deposits

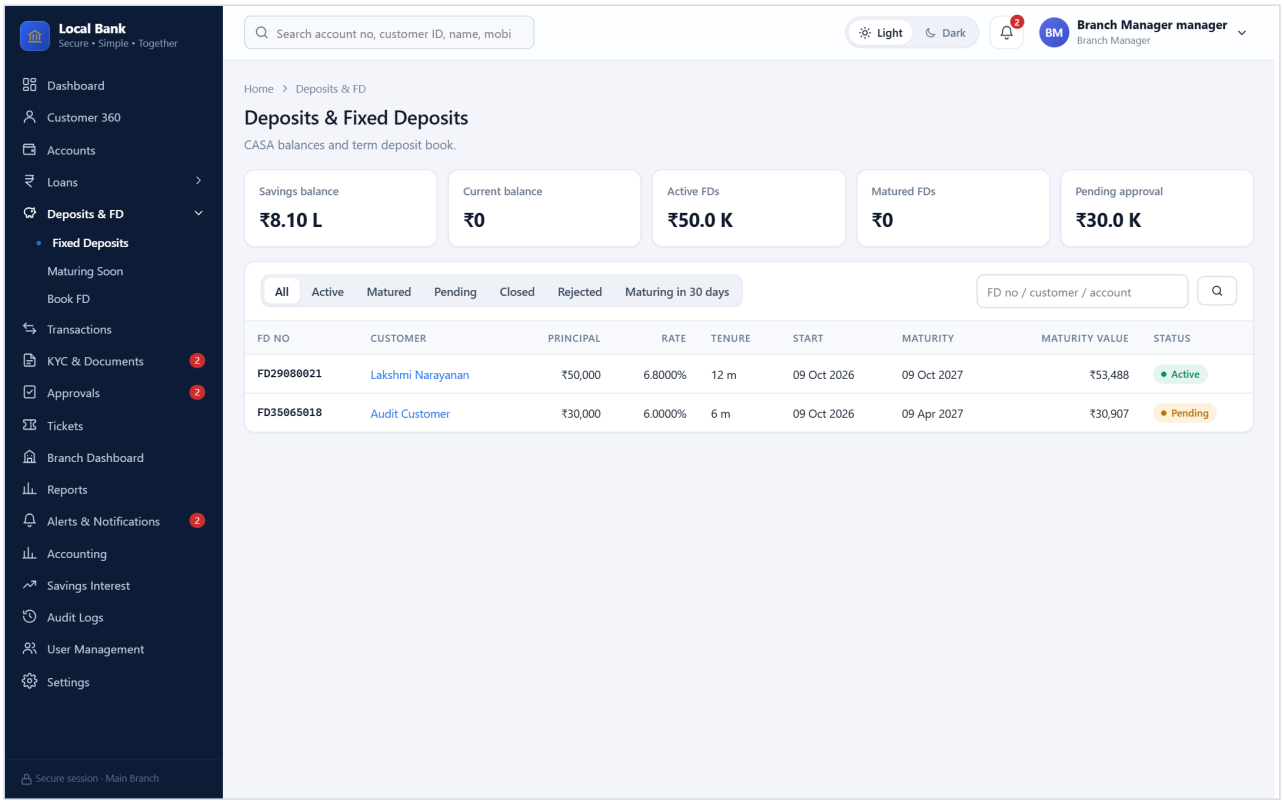



*Figure 9.1 — Fixed deposits, with a 'Maturing soon' view in the menu.*

## Interest rates (default)

| Tenure | 6 months | 12 months | 24 months | 36 months | 60 months |
|---|---|---|---|---|---|
| Rate a year | 6.00% | 6.80% | 7.00% | 7.10% | 6.50% |

Interest compounds quarterly. The maturity amount is shown when the FD is booked.

## Booking an FD

1. Customer 360 → **Book fixed deposit**.
2. Choose the funding account, enter the amount (from ₹1,000, in rupees and paise) and choose the tenure.
3. Select **Book FD**. The confirmation states the amount, tenure and rate, the customer and the account to be debited.
4. The amount is debited **immediately** and the FD waits for approval. If it is rejected, the money is refunded to the account.

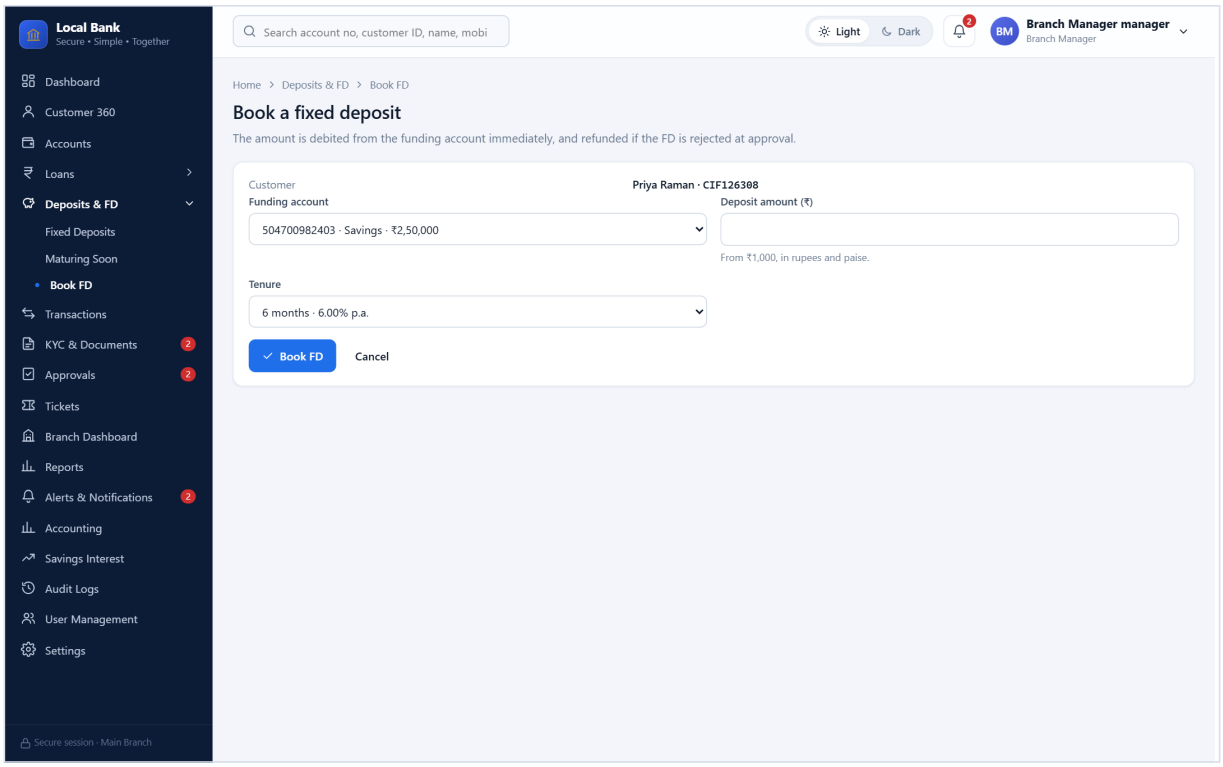



*Figure 9.2 — Booking a fixed deposit.*

## Maturity

On the maturity date the scheduled scan pays the maturity amount into the funding account automatically and marks the FD **Matured**. If no open account is linked, an alert asks staff to pay the customer manually.

## Closing an FD early

1. Customer 360 → **Deposits & FD** tab → **Close early**.
2. The screen shows the payout: interest is recalculated at the applicable rate **minus 1%** (default) for the time held.
3. Enter the reason and send the request. The money is paid out when the request is approved.

# 10. KYC and documents

## Uploading a document

1. Customer 360 → **Collect KYC document** (or KYC & Documents → **Upload Document**).
2. Choose the document type, enter the document number and, if it has one, the issue and expiry dates.
3. Attach the scan: **PDF, JPG or PNG**, up to **10 MB**. Files are checked for type and stored encrypted.
4. Select **Upload**. The document is **Pending** until another employee verifies it.

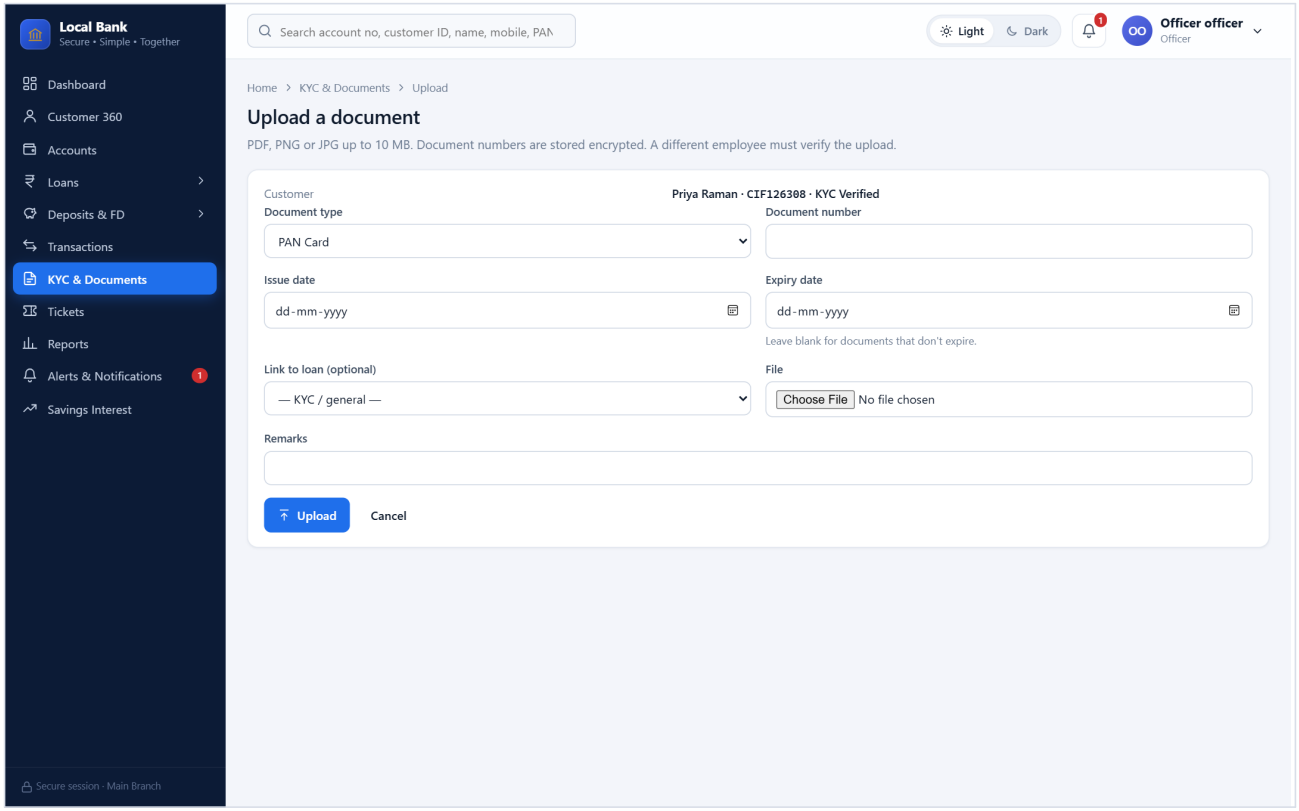



*Figure 10.1 — Uploading a KYC document.*

Document types: PAN Card, Aadhaar, Passport, Voter ID, Driving Licence, Address Proof, Photograph, Income Proof, Loan Agreement, Gold Valuation Certificate, Property Papers, Vehicle RC, Other.

# Verifying documents (checkers)

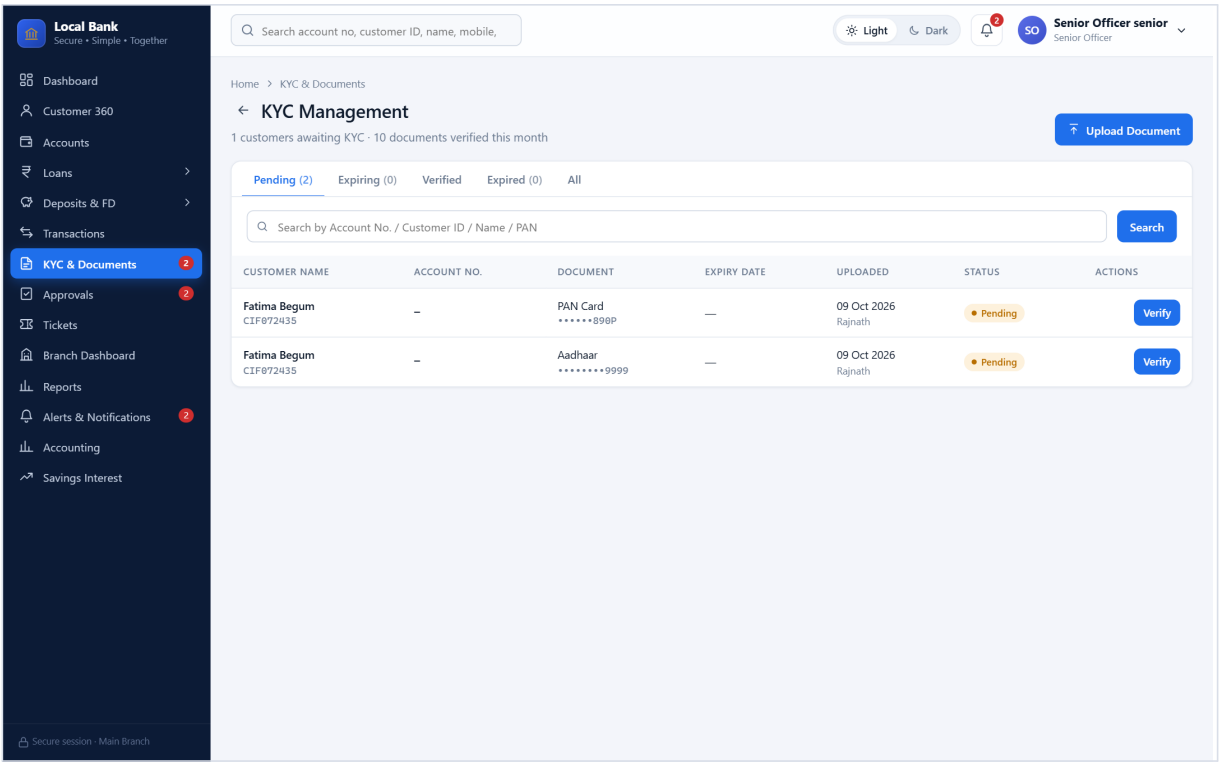



*Figure 10.2 — The KYC queue. The red number in the menu is how many documents wait for you.*

1. Open **KYC & Documents**. The **Pending** tab lists documents waiting for verification.
2. Select **Verify** on a document. Compare the original with the details and download the file if needed.
3. Select **Verify** if it is correct, or **Reject** and enter the reason (required — the customer must be told what to bring).
4. The application opens the **next pending document** automatically. When none are left, it says so.

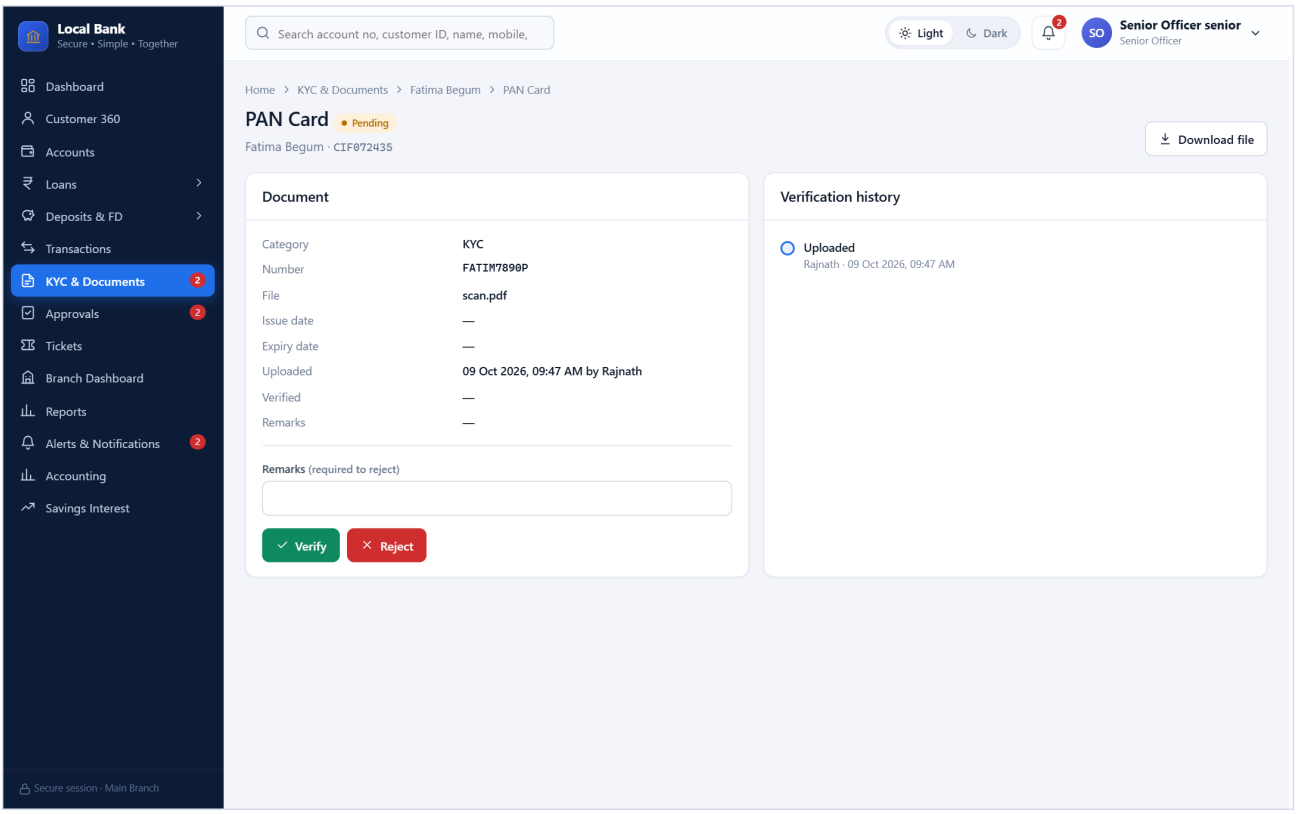



*Figure 10.3 — Document page with Verify and Reject.*

> **Tip:** You cannot verify a document you uploaded yourself (maker-checker). A decided document cannot be decided again.

## How the customer's KYC status is worked out

| Customer KYC | When |
|---|---|
| Verified | PAN Card and Aadhaar are both verified (and no current document has expired). |
| Pending | Documents are missing or still waiting for verification. |
| Rejected | A current KYC document was rejected. |
| Expired | A current KYC document has passed its expiry date. Upload and verify a renewed document to clear it. |

The **Expiring** tab lists documents expiring within 30 days (default) so customers can be contacted in time.

# 11. Approvals (maker-checker)

Sensitive actions are carried out in three steps by different people:

| Step | Who | What they do |
|---|---|---|
| 1. Maker | The employee who raises the request | Fills in the form. Nothing changes yet. |
| 2. Checker | Senior Officer or above (not the maker) | Reviews the request and forwards it, or rejects it. |
| 3. Approver | Branch Manager or above (not the maker or checker) | Approves — the change happens immediately — or rejects it. |

Two request types need a head-office approver (level 90 or above) instead of a Branch Manager: **Savings Interest Posting** (by default only the Super Admin holds interest approval) and **Role Permission Change** (Bank Admin or Super Admin).

## Request types

| Request | Raised from |
|---|---|
| Account Opening / Closure / Freeze / Unfreeze | Customer 360 → Accounts |
| Loan Sanction, Loan Interest-rate Change | New loan application; loan page |
| Fixed Deposit Booking, FD Premature Closure | Book FD; Deposits tab → Close early |
| High-value Transaction, Fund Transfer | Post transaction / Transfer at or above the limit |
| Transaction Reversal | Reverse button on a transaction |
| Customer Details Change | Customer 360 → Edit |
| Savings Interest Posting | Accounting → Savings Interest |
| Role Permission Change | User Management → Roles (when not made by the Super Admin) |

# Working your approval queue

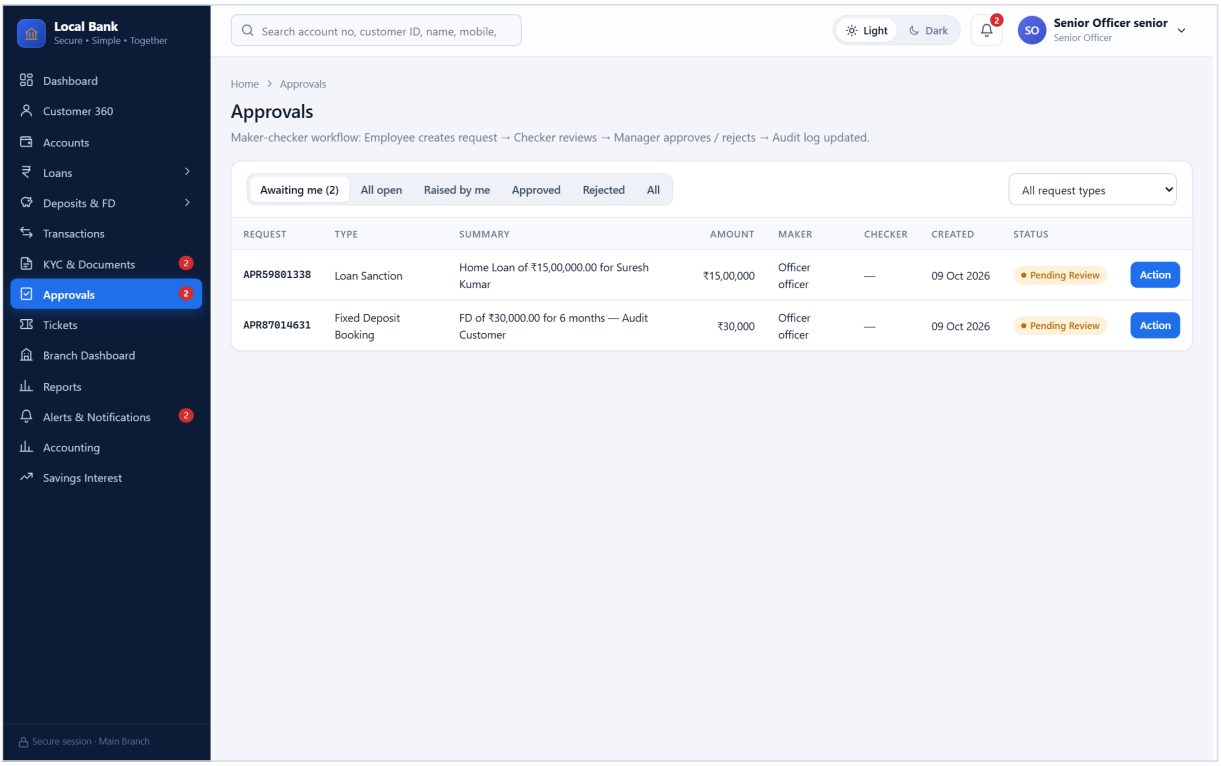



*Figure 11.1 — 'Awaiting me' lists only the requests you can act on, oldest first.*

1. Select **Approvals** in the menu (the red number is how many wait for you). The **Awaiting me** tab opens.
2. Select **Action** on the first request.
3. Read the request details: customer, amount, maker and the changes requested. Use the links to open the customer or loan.
4. Type remarks if needed, then select **Review & forward** (checker) or **Approve** (approver). The confirmation names the request and summarises it.
5. To refuse, select **Reject** and enter the reason (required).
6. The application moves to the **next request in your queue** automatically. When it is empty you are told so.

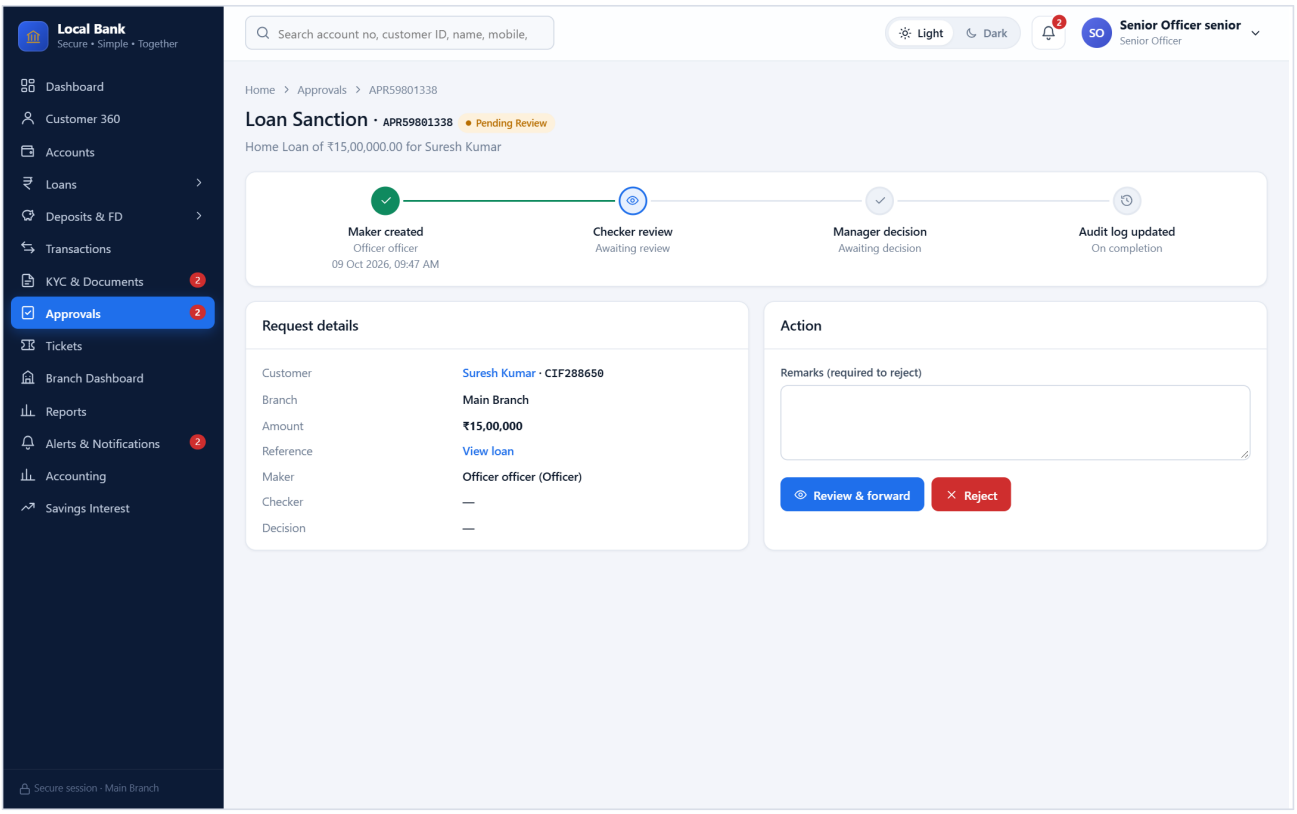



*Figure 11.2 — A request with its progress, details and actions.*

> **Tip:** If an approval is refused with a message (for example, KYC lapsed while the request waited), nothing was changed. Resolve the problem and approve again, or reject the request.

# 12. Customer tickets

Use tickets for complaints and service requests (card blocked, cheque book not received, charges disputed…).

1. Customer 360 → **Log complaint / service request**.
2. Enter a subject, category (Cards, Charges, Profile, Services, Digital, Loans, Deposits, Other), priority and description.
3. Open the ticket later from **Tickets** to assign it, add notes and change its status.

Statuses: **Open** → **Assigned** → **In Progress** → **Pending** → **Resolved** → **Closed**. Every change and note is kept in the ticket history.

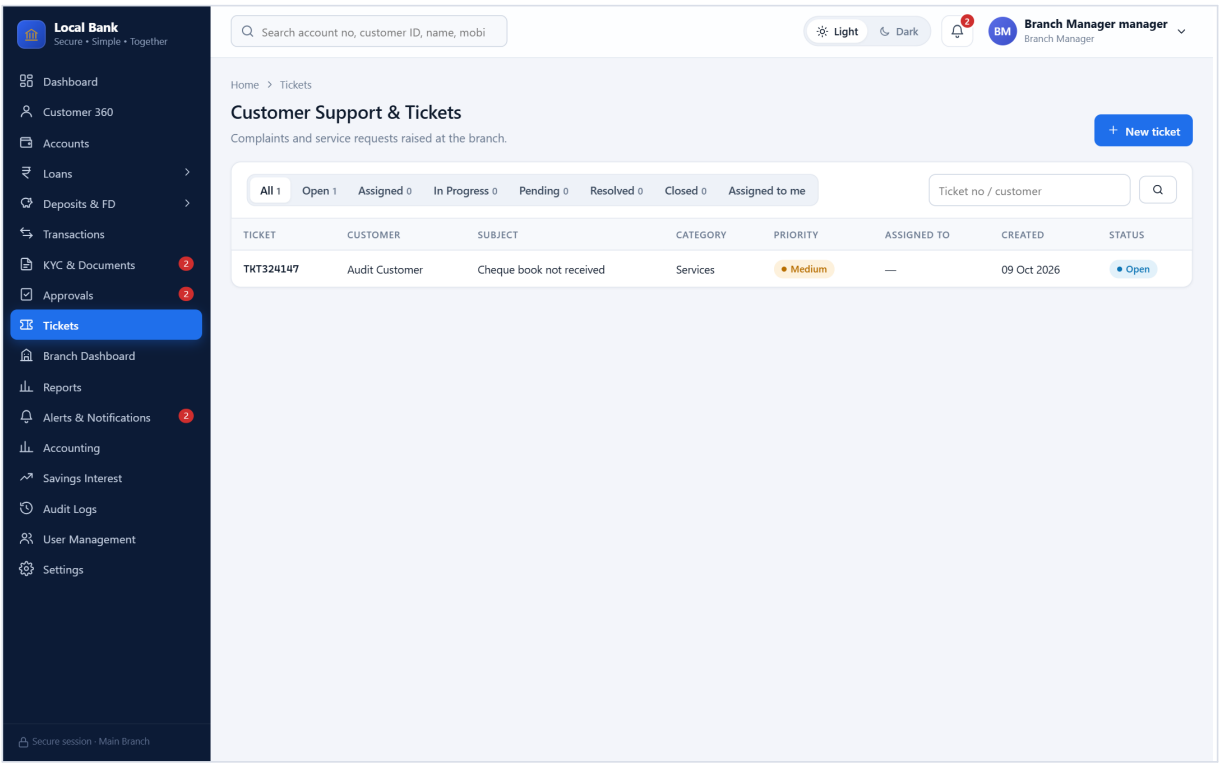



*Figure 12.1 — Ticket list.*

# 13. Alerts and notifications

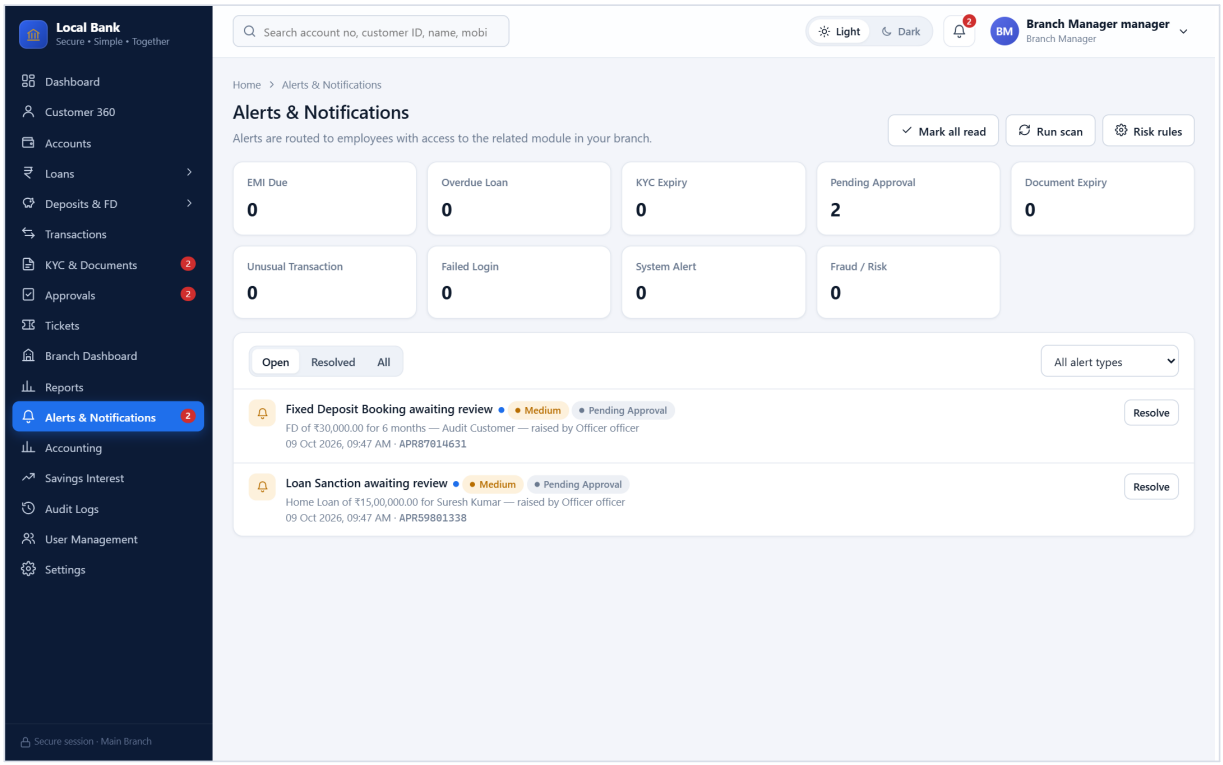



*Figure 13.1 — Alerts, by severity and type.*

Alerts are raised automatically and go to the employees whose role covers that area, in the affected branch:

- **Fraud & risk** — high-value transactions, rapid transactions, activity on dormant accounts.
- **Loans** — EMI due soon, overdue loans.
- **KYC** — documents expiring or expired.
- **Deposits** — FDs maturing; payout problems.
- **Security** — repeated failed sign-ins, sign-in from a new network.
- **Approvals** — requests waiting for action.

Select an alert to open the related record (this marks it read). Select **Resolve** when it has been dealt with, or **Mark all read**. Administrators can change thresholds, severity and routing under **Risk rules**.

# 14. Reports

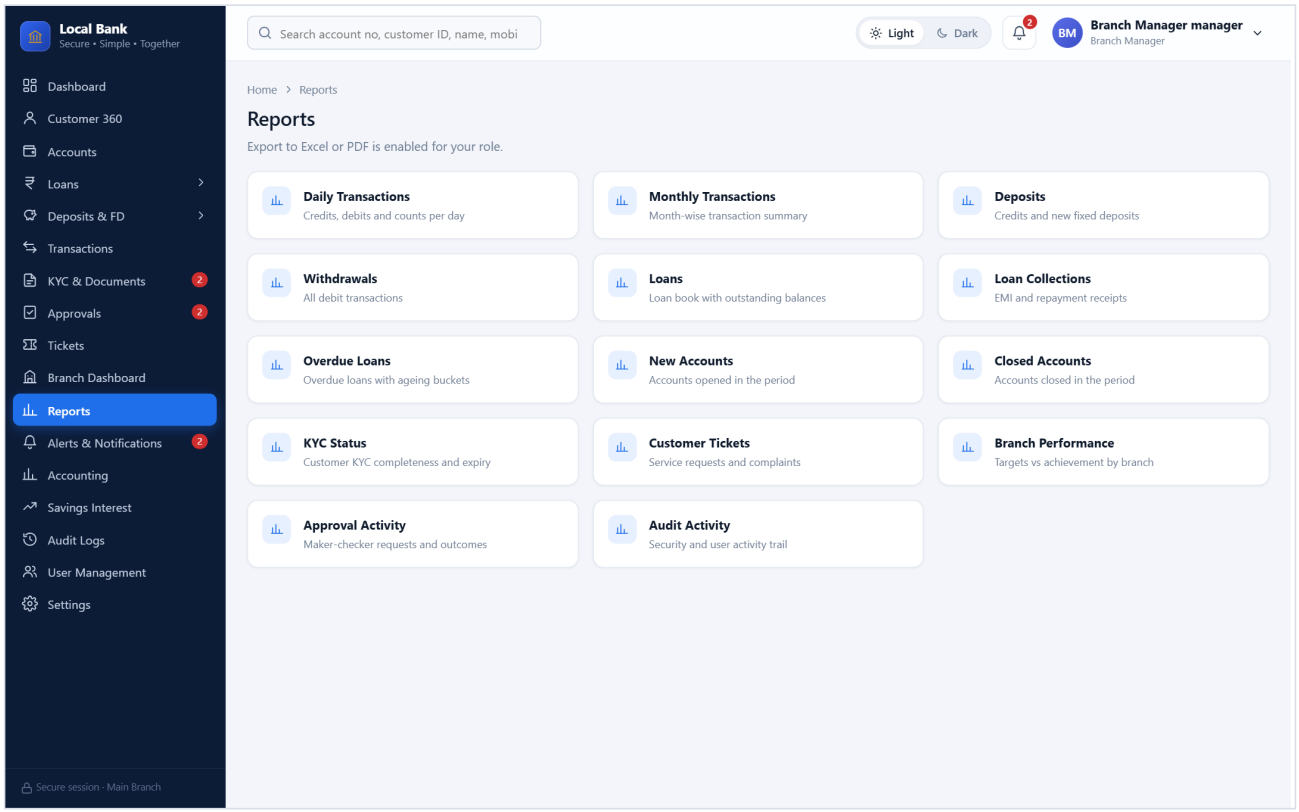



*Figure 14.1 — Available reports.*

| Report | Contents |
|---|---|
| Daily Transactions | Credits, debits and counts per day |
| Monthly Transactions | Month-wise transaction summary |
| Deposits | Credits and new fixed deposits |
| Withdrawals | All debit transactions |
| Loans | Loan book with outstanding balances |
| Loan Collections | EMI and repayment receipts |
| Overdue Loans | Overdue loans with ageing buckets |
| New Accounts | Accounts opened in the period |
| Closed Accounts | Accounts closed in the period |
| KYC Status | Customer KYC completeness and expiry |
| Customer Tickets | Service requests and complaints |
| Branch Performance | Targets vs achievement by branch |
| Approval Activity | Maker-checker requests and outcomes |
| Audit Activity | Security and user activity trail |

1. Open **Reports** and select a report.
2. Choose the period (and branch, for head-office roles).
3. Read it on screen, or select **Excel** or **PDF** to download it.

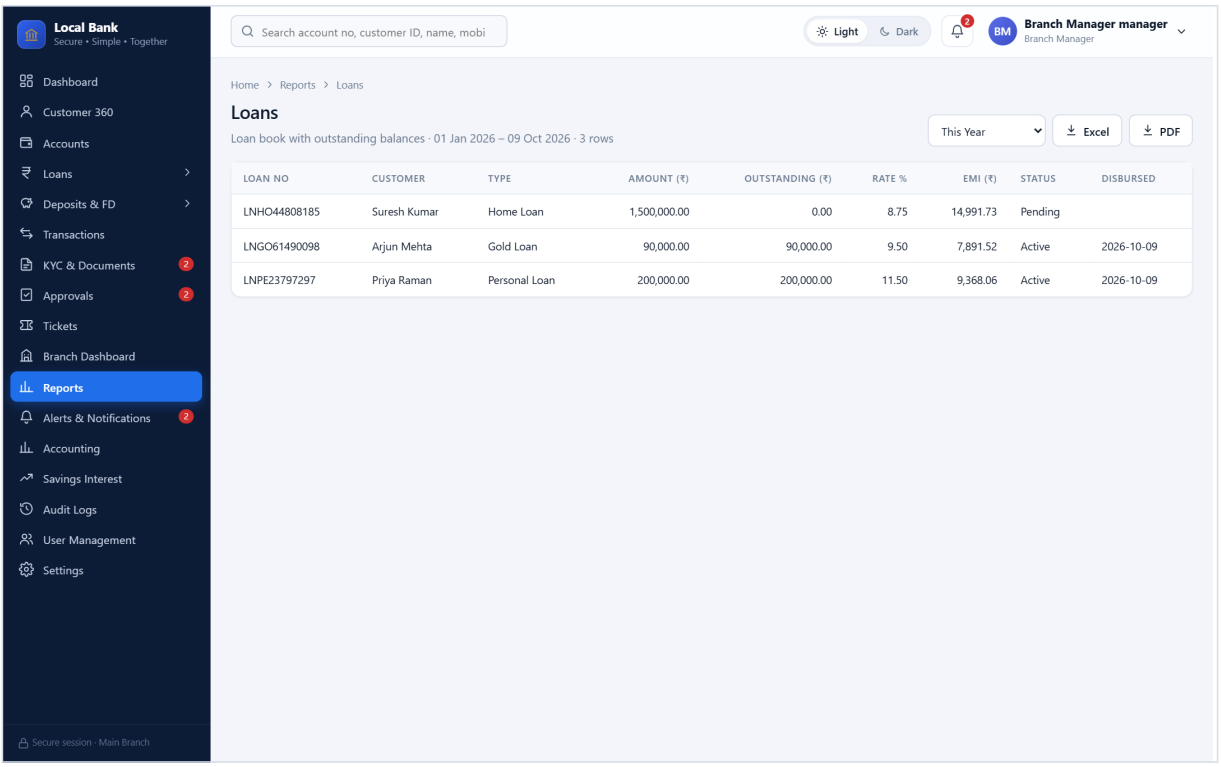



*Figure 14.2 — A report with export buttons.*

- Exporting needs the *export* right for that area. Each export is recorded in the audit log.
- You can export up to **30 times an hour**, and up to **50,000 rows** per export (default). Narrow the period for bigger reports.
- Exported files contain customer data. Store them only on approved locations and delete them when no longer needed.

# 15. Accounting and savings interest

Every money movement is posted to a **double-entry ledger**: each entry has equal debits and credits, and entries can never be edited or deleted.

## Accounting overview

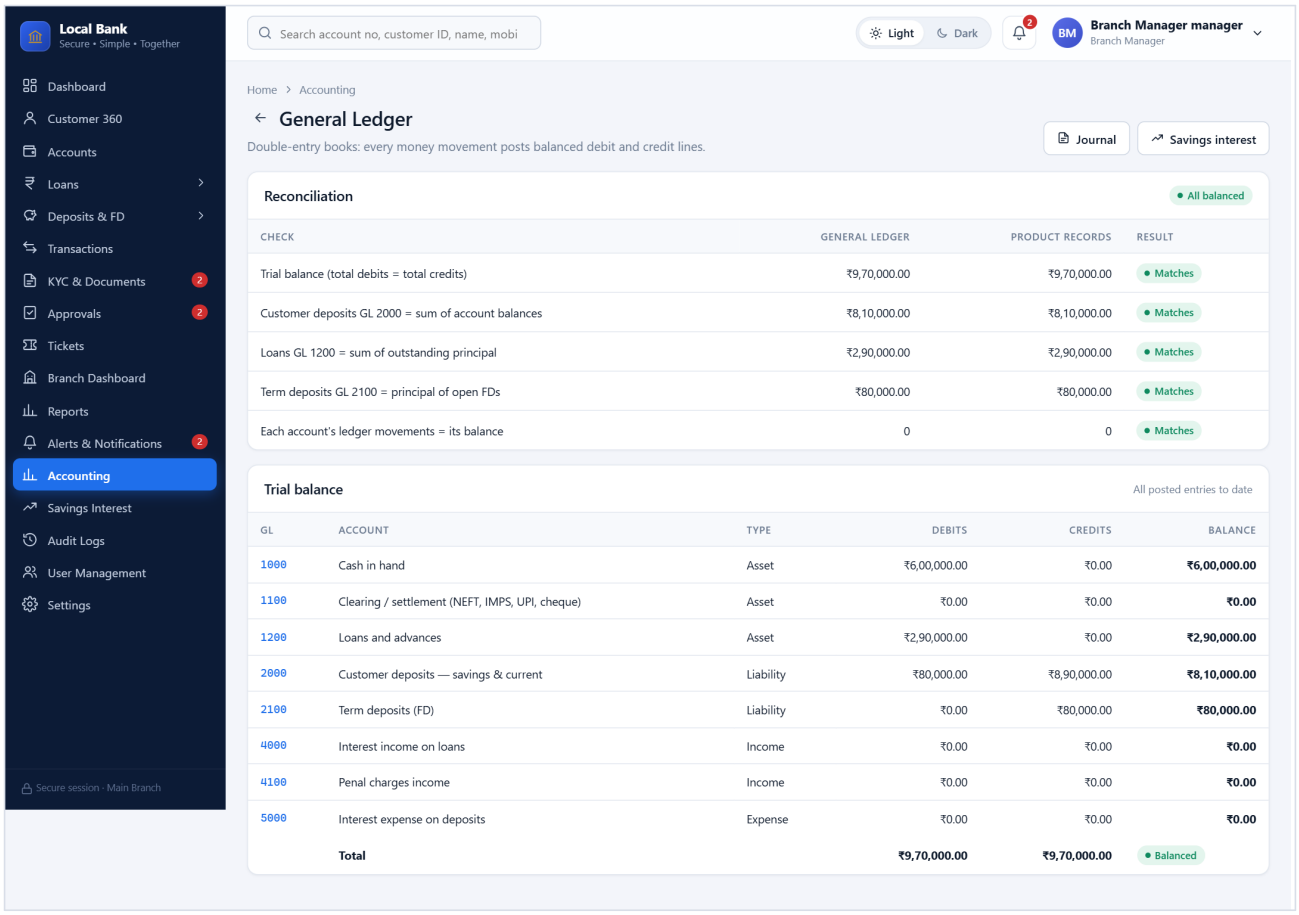



*Figure 15.1 — Trial balance and reconciliation checks.*

- **Trial balance** — balance of each ledger account (cash, customer deposits, term deposits, loans, interest income and expense).
- **Reconciliation** — checks that the ledger agrees with the books: deposits equal account balances, loans equal outstanding principal, term deposits equal open FDs, and each account's ledger equals its balance. All should show **OK**.
- **Journal** — every entry, searchable by ledger account or reference.

> **Important:** If any reconciliation check shows a difference, stop and inform the Branch Manager and head office immediately.

## Posting savings interest

1. Open **Savings Interest**, choose the period (usually a quarter) and select **Recalculate preview**.
2. Check the interest per account. It is calculated on each day's closing balance at the account's rate (actual/365).
3. Select **Submit for approval**. A Branch Manager reviews it; the Super Admin gives final approval and it posts.

4. The same branch and period can only be posted once.

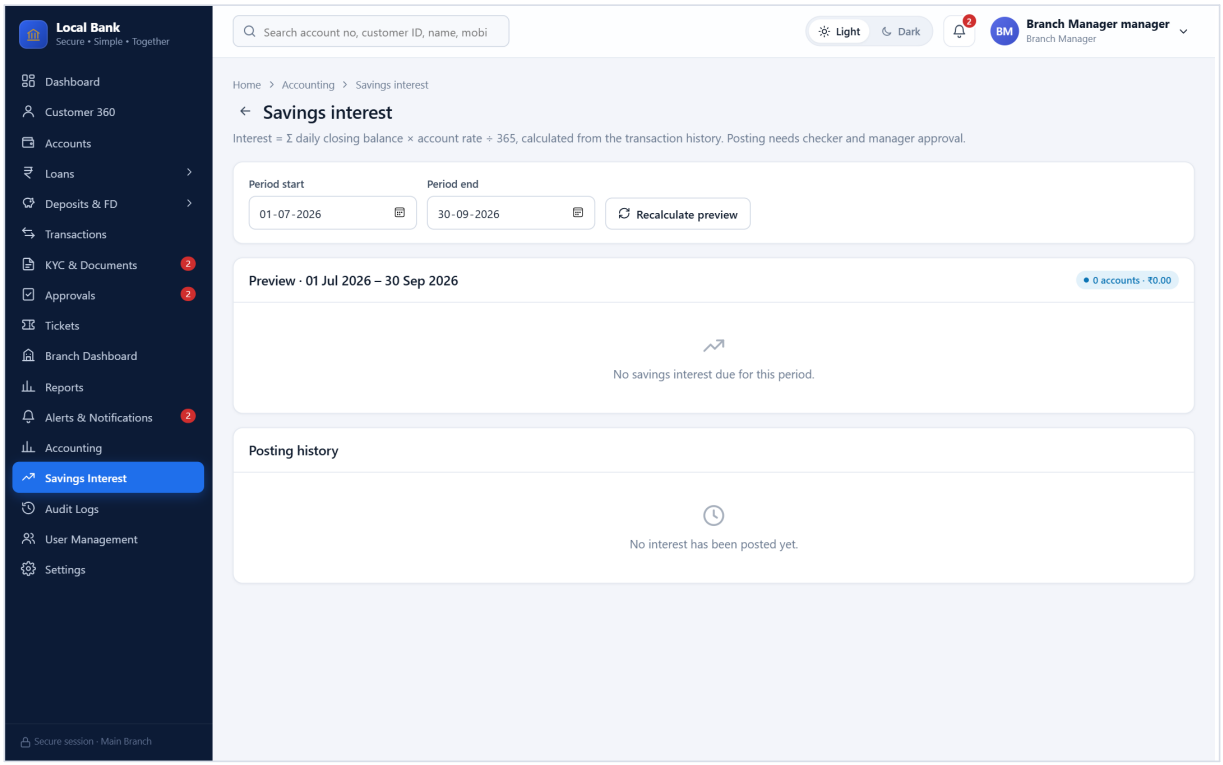



*Figure 15.2 — Savings interest preview.*

# 16. Administration

This chapter is for Super Admins, Bank Admins and Branch Managers.

## Adding an employee

1. **User Management → Add employee**.
2. Enter name and email, choose the role and branch.
3. Save. A one-time **invite link** is shown on screen (valid 48 hours). Give it to the employee in person or through an approved channel. It is not shown again.
4. The employee opens the link and sets their own password.

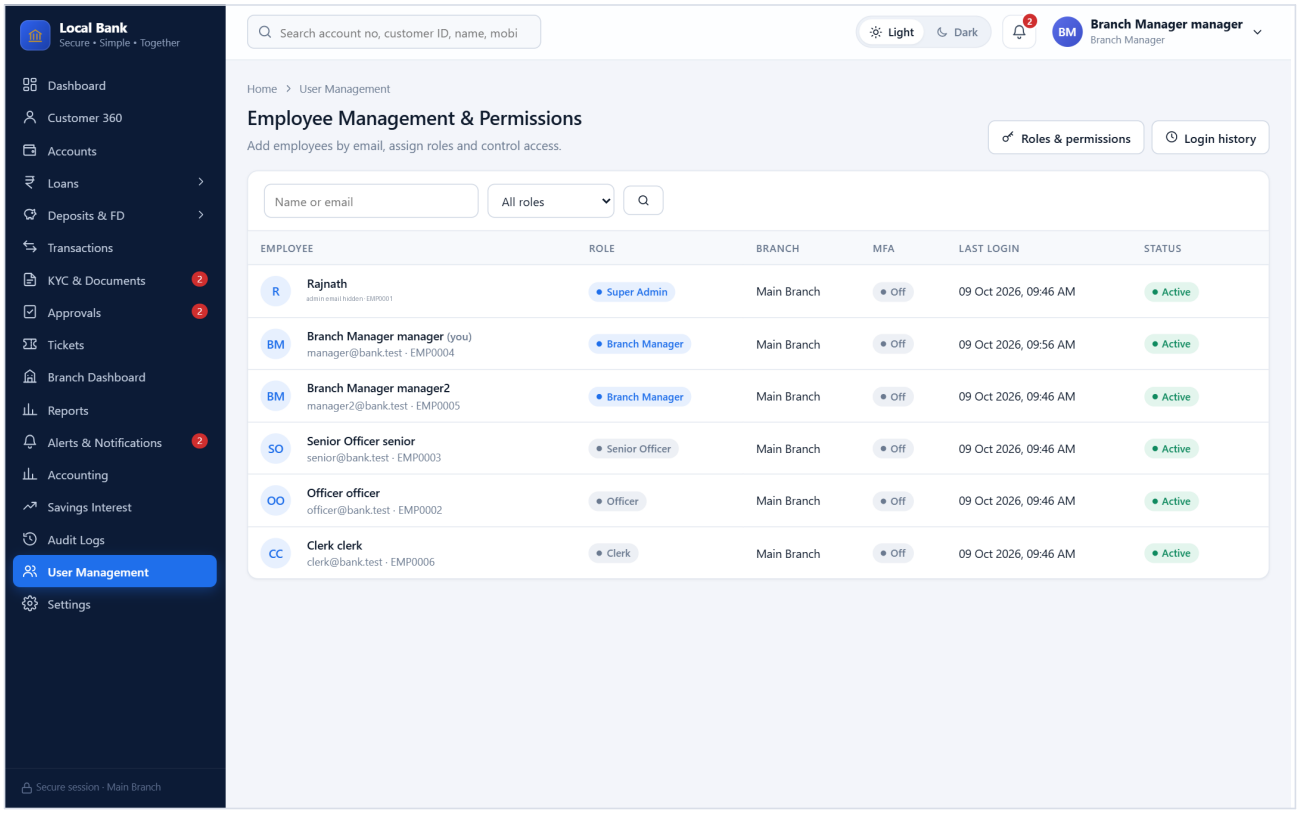



*Figure 16.1 — Employee list with role, branch, MFA and last sign-in.*

## Managing an employee

Open the employee from the list. You can change role or branch, disable the account, **Sign out all sessions**, **Unlock account** after too many wrong passwords, **Reset MFA**, or issue a **password reset link** (the old password stops working at once). You cannot manage yourself or someone with a higher role.

## Roles and permissions

Each role has rights per area: *view, create, edit, delete, approve, reject, export, print*. Open **Roles & permissions** to see or change them. Changes made by the Super Admin apply immediately; changes by anyone else become a **Role Permission Change** request for approval. The full matrix is in Appendix A.

# Settings

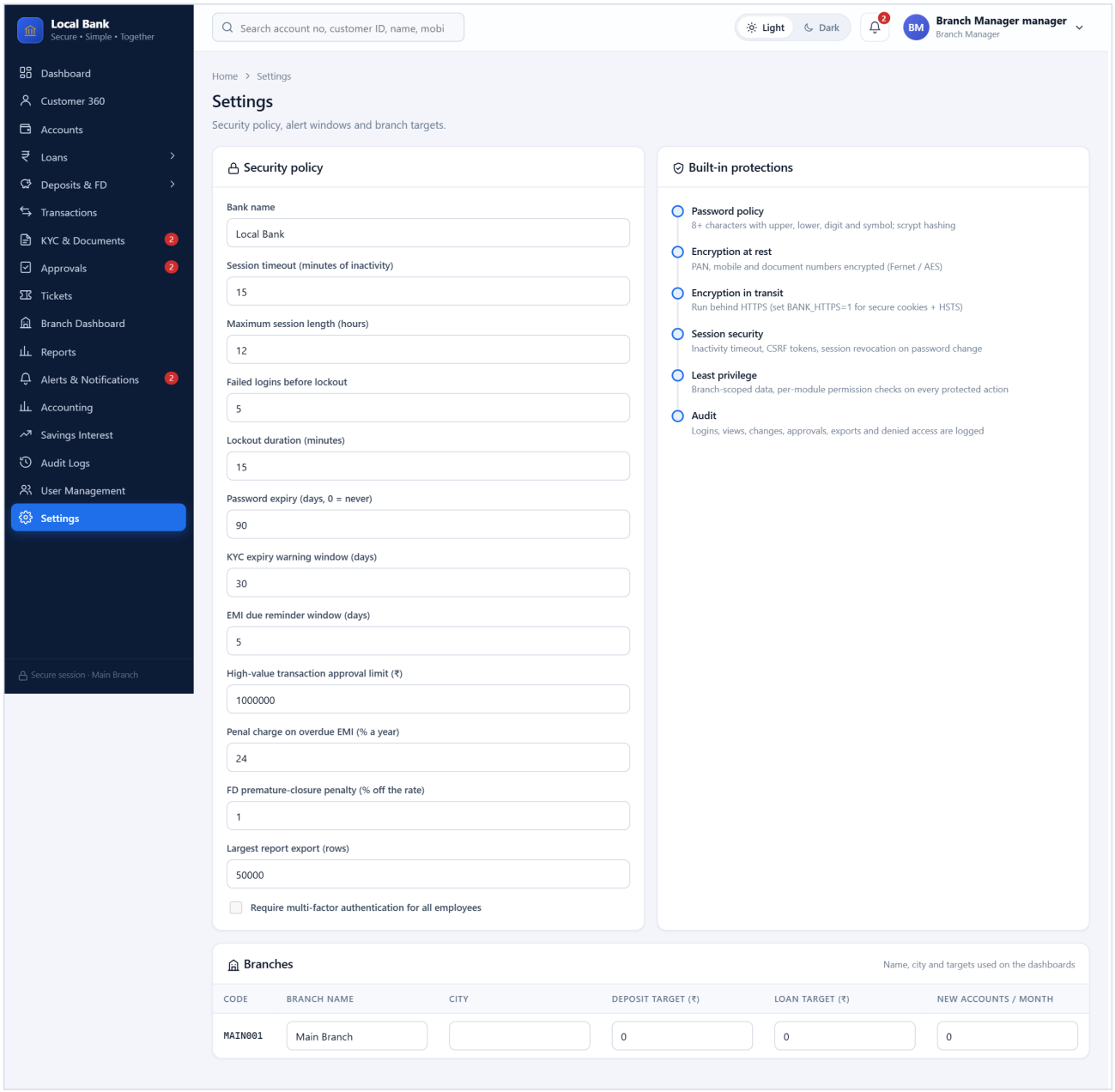



*Figure 16.2 — Settings (security policy, limits and branches).*

| Setting | Default | Meaning |
|---|---|---|
| Session timeout | 15 minutes | Sign out after this much inactivity |
| Maximum session | 12 hours | Sign out after this long regardless of activity |
| Failed sign-ins before lock | 5 | Then locked for the lockout period |
| Lockout period | 15 minutes | |
| Password expiry | 90 days | |
| Enforce two-factor | Off | When on, every employee must set up two-factor sign-in |
| KYC expiry warning | 30 days | When expiring documents are flagged |
| EMI due warning | 5 days | When EMI-due alerts are raised |
| High-value approval limit | ₹10,00,000 | Transactions at or above this need approval |

| Setting | Default | Meaning |
|---|---|---|
| Penal interest | 24% a year | On overdue instalments |
| FD premature penalty | 1% | Rate reduction on early closure |
| Maximum export rows | 50,000 | Per Excel/PDF export |

Branch Managers can edit their own branch only; opening a new branch needs a head-office administrator. Values outside the allowed range are refused.

## Audit logs

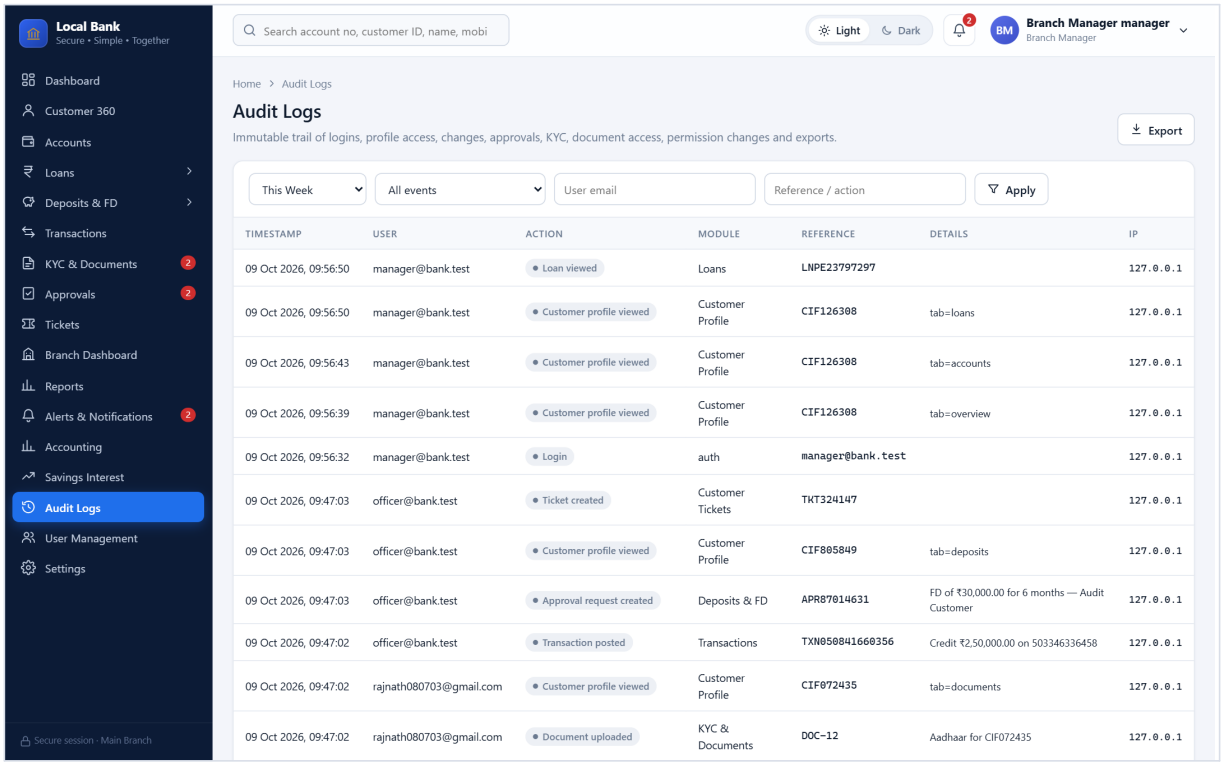



*Figure 16.3 — Audit log.*

Every sign-in, search, view of sensitive data, change, approval, export and denied attempt is recorded with the employee, time, IP address and result. Records are chained with a cryptographic hash, so any later change is detectable. They cannot be edited or deleted from the application.

## Running the application (technical)

| Task | How |
|---|---|
| Start the database | `docker compose up -d` |
| Start the application | `python run.py` → http://127.0.0.1:5000 (applies database updates first) |
| Check the books | `flask --app wsgi:app reconcile` |
| Check the audit chain | `flask --app wsgi:app verify-audit` |
| Run the scheduled scan | `flask --app wsgi:app scan` (overdue loans, KYC expiry, FD maturity, dormancy) |
| Back up / restore (server stack) | `deploy/backup.sh` · `deploy/restore.sh <file>` |
| Run all tests | `python tests/run_all.py` |

> **Important:** Settings and secrets are kept in the `.env` file on the server. Never share it, email it or copy it to a shared folder, and never put real passwords in screenshots or tickets.

# 17. Good practice and troubleshooting

## Security good practice

- Never share your password, invite link or two-factor codes — not even with colleagues or IT.
- Lock or sign out of your computer whenever you leave the counter.
- Reveal PAN or mobile numbers only when needed; every reveal is logged.
- Read every confirmation dialog before you confirm.
- Do not export reports to personal devices or email. Delete exported files when finished.
- Report anything unusual — an unexpected alert, a request you did not raise — to your manager straight away.

## Common questions

| Problem | What to do |
|---|---|
| “Invalid email or password” | Check Caps Lock. After 5 wrong attempts wait 15 minutes, or ask an administrator to unlock you. |
| “Too many attempts” (error 429) | Wait one minute and try again. |
| Signed out unexpectedly | Sessions end after 15 minutes idle or 12 hours in total. Sign in again; unsaved forms must be re-entered. |
| A menu item or button is missing | Your role does not include that right. Ask your manager if you need it. |
| “Not approved: …” on an approval | Nothing changed. The message explains why (e.g. KYC lapsed, balance not zero). Fix it first. |
| Cannot open an account or loan | The customer's KYC must be Verified. Check the Documents tab. |
| Cannot verify a document | You uploaded it yourself — another employee must verify it. |
| Withdrawal refused on a Frozen account | Correct: frozen accounts accept only deposits. An unfreeze needs approval. |
| Large deposit did not post | At or above ₹10,00,000 it goes for approval first; it posts when approved. |
| Posted the wrong amount | Use Reverse on the transaction (needs approval), then post the correct one. |
| Customer not found when opening an account | Type at least 3 letters of the name, or the exact mobile, PAN, customer ID or account number. |
| Export refused | You need the export right; you may have reached 30 exports/hour or the row limit — narrow the period. |
| “Something went wrong” page | Note the reference shown on the page and give it to your administrator; it identifies the exact error in the logs. |

## Using the application on a phone or tablet

The application adapts to small screens. Use the menu button at the top left to open the menu.

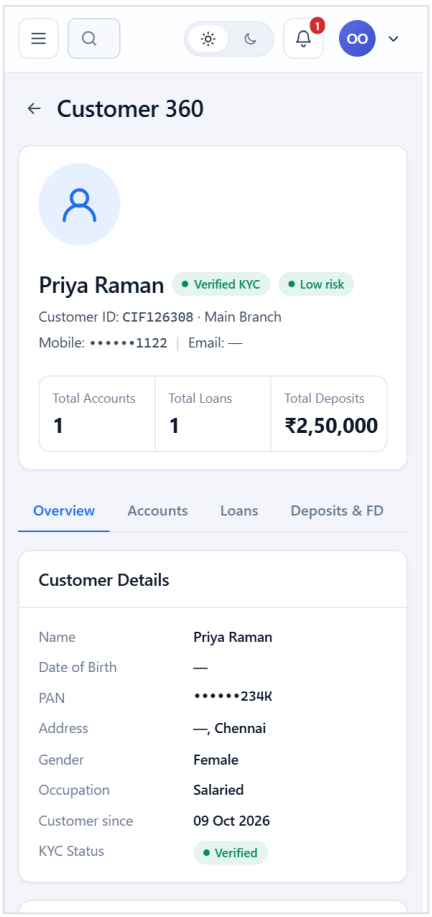



*Figure 17.1 — Customer 360 on a phone-sized screen.*

# 18. Appendix A — Roles and permissions

Default rights for each system role, taken from the application. V = view, C = create, E = edit, D = delete, A = approve, R = reject, X = export, P = print. Loans includes loan approval. **Admin*** = User Management, Settings and Audit Logs combined. Administrators can adjust these in Roles & permissions.

| Role | Level | Purpose |
|---|---|---|
| Super Admin | 100 | Full access to every module and branch. |
| Bank Admin | 90 | Head-office administrator: all branches, staff, roles and settings; no money movement. |
| Branch Manager | 80 | Runs a branch; final approver in maker-checker. |
| Senior Officer | 60 | Checker for loans, accounts, deposits and KYC. |
| Accountant | 55 | Books and reconciliation: ledger, trial balance, interest runs (maker). |
| Auditor | 50 | Internal / statutory auditor: read everything, export reports and audit logs, change nothing. |
| Loan Officer | 45 | Originates and services loans; maker for loan applications and repayments. |
| Officer | 40 | Maker for accounts, loans, deposits; handles customers. |
| Clerk | 20 | Front desk: lookups, cash transactions, document intake. |
| Read-Only | 10 | Look up customers, accounts, loans and deposits without changing anything. |

Columns: Cust. = Customer profile, Acct. = Accounts, Txn. = Transactions, FD = Deposits & FD, Rep. = Reports.

| Role | Cust. | Acct. | Txn. | Loans | FD | KYC | Rep. | Admin* |
|---|---|---|---|---|---|---|---|---|
| Super Admin | All | All | All | All | All | All | All | All |
| Bank Admin | V | V | V | V | V | V | VXP | VCEDARX |
| Branch Manager | VCEARX | VCEARXP | VCARXP | VCEARXP | VCEARXP | VCEARX | VXP | VX |
| Senior Officer | VCEAR | VCEARP | VCARP | VCEARP | VCEARP | VCEAR | VXP | — |
| Accountant | — | V | VXP | V | V | — | VXP | — |
| Auditor | V | V | VX | V | V | V | VX | VX |
| Loan Officer | V | V | V | VCEP | — | VC | V | — |
| Officer | VC | VCP | VCP | VC | VC | VC | V | — |
| Clerk | — | V | VC | — | — | VC | — | — |
| Read-Only | V | V | V | V | V | V | — | — |

# 19. Appendix B — Glossary

| Term | Meaning |
|---|---|
| Customer 360 | One screen with everything about a customer. |
| CIF / Customer ID | The customer's unique number, e.g. CIF392382. |
| KYC | Know Your Customer: verified identity and address documents. |
| Maker-checker | Two-person control: one employee makes a request, others review and approve it. |
| EMI | Equated monthly instalment of a loan. |
| Penal charge | Extra charge on an overdue instalment for the days it is late. |
| Payoff | Amount that closes a loan today: principal + interest + penal charges. |
| LTV | Loan-to-value: loan amount as a percentage of the gold valuation (maximum 75%). |
| FD | Fixed deposit for a fixed tenure at a fixed rate. |
| Premature closure | Closing an FD before maturity, at a reduced rate. |
| Dormant account | No customer transaction for 365 days. |
| Frozen account | Debits blocked; credits allowed. |
| Trial balance | List of ledger balances; total debits must equal total credits. |
| Reconciliation | Checking that the ledger agrees with account, loan and FD balances. |
| Audit log | Tamper-evident record of who did what and when. |
| Two-factor authentication (MFA) | Sign-in with a password plus a code from an authenticator app. |